Occupation Ladders over the Business Cycle*

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Abstract

This paper studies occupational mobility over the business cycle. We divide occupations into two broad groups, “attractive” and “nonattractive,” where we label an occupation attractive if the total net inflow into this occupation through job-to-job transition is positive. We measure total net inflows into both occupation groups. We measure these inflows separately for job-to-job transitions and transitions from unemployment to employment. We find that the net inflow from nonattractive to attractive occupations through job-to-job transitions slows during recessions. The net inflow through transitions from unemployment has a similar cyclicality. This finding suggests a novel cost of recession: during recessions, workers have fewer opportunities to move to a better occupation.

Keywords: occupational mobility, business cycles, worker flows

JEL Classification: E24, E32, J24, J62

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1 Introduction

Measuring the cost of business cycles has been an important research topic in macroeconomics. The cost of business cycles has been studied in various contexts. The original Lucas (1987) calculation analyzed the cost of consumption variation. Since then, different researchers have considered different aspects of recessions. Examples include the increase in unemployment,[1] which can also lead to a loss of human capital,[2] a decline in firm entry,[3] which can harm future employment growth and also innovation,[4] and a decline in job-to-job transition, which creates misallocation of talents across jobs.[5]

Occupations are viewed as one of the most important attributes when we analyze an individual's labor market situation. An earlier work of Kambourov and Manovskii (2009) emphasizes the occupation specificity of human capital, which determines the wages and workers' career paths. Studies utilizing the "task approach" (e.g., Acemoglu and Autor, 2011) characterize an occupation as the combination of various tasks and analyze how changes in the economic environment affect each occupation. Recent applied micro studies, such as Yamaguchi (2012), Guvenen et al. (2020), and Lise and Postel-Vinay (2020), reveal rich interactions between workers' skills and heterogeneous occupations.

This paper provides a novel perspective on the cost of recessions through the lens of occupations. We show the workers continuously switch to different occupations, and the movement to "better" occupations, which we call the *occupation ladder*, slows during recessions. This result indicates the recession reduces the opportunities for workers to climb the occupation ladder.

Applying the insight of Sorkin (2018) to the context of occupation switch, we divide the occupations into two groups: "attractive" occupations and "nonattractive" occupations. The division is based on the net flows of workers across occupations when they make a job-to-job transition.

Based on this division of occupations into two groups, we analyze the cyclicality of

[1] See, for example, Mukoyama and Şahin (2006) and Krusell et al. (2009).
[2] See Krebs (2007).
[3] See Lee and Mukoyama (2015).
[4] See Barlevy (2007) and Sedláček and Sterk (2017).
[5] See Baydur and Mukoyama (2020).

the net flows across occupation groups. We focus on the occupation switch upon job-to-job transitions and when a worker finds a new job from unemployment. We find that, during recessions, the net flow of climbing the occupation ladder declines for both job-to-job transitions and unemployment-to-employment transitions. The cyclical movement of separation and vacancies for these occupation groups is consistent with the interpretation that the value of attractive occupations declines relatively more than that of nonattractive occupations during recessions. These facts point to the view that, during recessions, workers have reduced access to opportunities to move up the occupation ladder.

To measure the cost of the slower climb of the occupation ladder during recessions, we construct a simple model where workers face stochastic shocks of job finding, job separation, job-to-job transitions, and occupation switching. The calibrated model shows that approximately one-fifth of the total cost of recessions is due to the slowdown in occupation switching. Therefore, this novel cost constitutes an economically significant part of the cost of recessions.

This paper is related to several strands of literature. The first is the "cost of business cycles" literature listed above, starting from Lucas (1987). Various studies in this literature argue that recessions impose various economic costs on consumers. Our work suggests a novel source of the cost of recessions to the economy: slowing down the occupation ladder.

Another related literature is the recent analysis of job-to-job flows. Workers are known to, on average, move to higher-paying jobs when they experience job-to-job transitions, and this movement is an important component of wage growth over the life cycle (see, e.g., Topel and Ward, 1992). The movement of workers toward better jobs over time is often referred to as the "job ladder." The job-to-job transition rate is highly cyclical (see, for example, Fallick and Fleischman, 2004; Hyatt and McEntarfer, 2012; Haltiwanger et al., 2018; Moscarini and Postel-Vinay, 2018), and the cyclical nature of the job-to-job transition affects how microeconomic match characteristics are distributed in the economy (Gertler et al., 2020; Baydur and Mukoyama, 2020). Studies such as Barlevy (2002) and Mukoyama (2014) show that fluctuations in job-to-job flows associated with business cycles can have a macroeconomic impact. Our paper can be viewed as an occupation analog of the job-ladder analysis in this literature. Moreover, because an occupational switch often occurs together with a job-to-job transition, the cyclical movement of the job-to-job transition rate interacts nontrivially with the occupation ladder.

Some earlier works have analyzed the patterns of occupational mobility. Moscarini and Thomsson (2007) document the occupational mobility in the US. Moscarini and Vella (2008) analyze the mobility pattern over the business cycle. Neither paper considers the "ladder" aspect of the mobility—they do not take a stand on which occupation is better or worse. Naturally, their interest is gross flow—how often people move occupations and why. Our primary focus is the net flow on the ladder—how the tendency to move up to a better occupation interacts with the business cycle. Another difference is that we focus on occupational switches upon job-to-job transitions and unemployment-to-employment transitions. These earlier papers consider all gross movements, including the occupational switch within the same employer.

Several earlier and contemporaneous papers analyze concepts similar to our occupation ladder. Evans (1999) ranks occupations using a social prestige survey, and both occupation upgrading and downgrading are procyclical in British data. Devereux (2002) ranks the occupations by wages and finds evidence for cyclical upgrading in the US. More recently, Cortes et al. (2024) similarly measure the occupational ranking ("occupational ladder" in their terminology) by average wages and analyze its role in life cycle wage growth. Carrillo-Tudela et al. (2025) represent occupation as a task portfolio, and analyze how this portfolio changes over the business cycle. Carrillo-Tudela and Visschers (2023) emphasize the cyclicality of occupational switching upon unemployment-to-employment transitions as a factor that influences unemployment dynamics. They do not explicitly consider the occupation ladder, and their focus is on positive analysis, rather than normative analysis, which is our focus. Carrillo-Tudela et al. (2022) analyze occupation switching through job-to-job transitions. Their main focus is earnings dynamics, but they also conduct "cost of business cycles" calculation. Their analysis is based on task-based categorization of occupations (and earnings change upon switching). All the above papers rank occupations differently from the current work, and from that perspective, they are complementary to our study.

This paper is organized as follows. In Section 2, we set up the data and define two categories of occupation: "attractive" and "nonattractive." In Section 3, we examine the patterns of net flows across occupations. In Section 4, we analyze additional statistics that help us understand the mechanism underlying the cyclical patterns of net flows. In Section 5, we look at another dataset. Section 6 builds a model to investigate the implications of the cyclical patterns of occupational flows on the cost of recessions. Section 7 concludes.

2 Baseline data and definitions

We use micro-level data from the Current Population Survey (CPS) to quantify occupational mobility. We obtained monthly CPS data from the Integrated Public Use Microdata Series (IPUMS-CPS) database (Flood et al., 2020). CPS uses rotation groups in which each individual is interviewed for four consecutive months, not interviewed for eight months, and then re-interviewed for another four months before leaving the sample. The rotating panel design of CPS enables us to observe each individual's employment status next month. In every round, unemployed individuals report their occupation at their latest employer, and employed individuals report their occupation in their current job. After the redesign in 1994, employed individuals also report whether they work for the same employer they reported in the previous month. Using this information, we construct occupational mobility measures separately for those who make a job-to-job transition, EE, and a transition to a new employer with an intervening unemployment spell, UE. Our final sample covers the period from September 1995 to December 2018.

A challenging issue with measuring occupational mobility is the changes in the coding scheme used to record individuals' occupations in the original CPS data. CPS uses the Census classification system for occupations. For the period we cover, one major change occurred to the occupational coding scheme starting in 2003, and a minor change occurred starting in 2011. To alleviate possible measurement errors, we use the IPUMS-CPS variable "OCC2010," which provides a harmonized occupation variable for all the survey months mapped to the Census 2010 occupational classification. We note that the 2010 Census classification scheme and the corresponding IPUMS-CPS variable are based on the 2010 Standard Occupation Classification

Occupation title	Net inflow rates via EE	Log-Deviation from Median Wage	Task Intensity
Management	6.479	0.980	NRC
Business and financial operations	7.718	0.569	NRC
Computer and mathematical	6.576	0.777	NRC
Architecture and engineering	9.844	0.719	NRC
Life, physical, and social science	21.838	0.556	NRC
Community and social services	12.371	0.163	NRC
Legal	−0.653	0.801	NRC
Education, training, and library	−6.286	0.302	NRC
Arts, design, entertainment, sports, and media	4.575	0.232	NRC
Healthcare practitioners and technical	4.009	0.536	NRC
Healthcare support	16.456	−0.285	NRM
Protective service	−1.156	0.053	NRM
Food preparation and serving related	−32.974	−0.615	NRM
Building and grounds cleaning and maintenance	−9.725	−0.404	NRM
Personal care and service	−6.125	−0.466	NRM
Sales and related	−20.624	−0.272	RC
Office and administrative support	7.061	−0.079	RC
Farming, fishing, and forestry	−9.084	−0.527	N/A
Construction and extraction	8.860	0.154	RM
Installation, maintenance, and repair	9.747	0.189	RM
Production	12.094	−0.105	RM
Transportation and material moving	2.200	−0.184	NRM

Table 1: Two-digit occupational categories

(SOC) scheme. The major occupation categories in the Census classification correspond to two-digit occupations in the SOC scheme. These occupations are listed in Table 1.

We further divide the two-digit occupations into two broader categories, *attractive* (denoted by A for "attractive") and *nonattractive* (denoted by N for "nonattractive"). We define an occupation i as attractive if the net inflow rate into i through an EE transition is positive, i.e., the occupation "attracts" workers. We calculate the total net inflow rate as follows. Let the total number of workers who experience the i-to-j occupational switch when they switch jobs be EE_{ij}. Then, the net inflow rate into occupation i is:

$$F_{EE,i} \equiv \frac{\sum_{j \neq i} EE_{ji} - EE_{ji}}{E_i}, \tag{1}$$

where the denominator E_i is the total employment in occupation i.

The net inflow rates for each occupation are displayed in the second column of Table 1. To avoid classification errors due to changes in the coding schemes, we restrict our sample in this section to observations after the 2002 changes were implemented when determining net inflows into occupation groups.[6] The total quantities above are observed monthly and allow for the construction of monthly net inflow rates. The calculations in Table 1 pool all observations in our sample, yielding results equivalent to an employment-weighted average of monthly net inflow rates. We analyze the business cycle patterns in the next section.[7]

Our classification based on net inflow rates via EE transitions is in the spirit of Sorkin (2018): by the revealed-preference argument, a large outflow and a small inflow through EE transitions imply people prefer to be in another occupation than in occupation i. Upon receiving another job offer, the worker has a choice of staying or moving. The fact that the worker moved to a new position implies that the worker prefers the new position. Aggregating the choices of many workers who also face idiosyncratic shocks, patterns of the worker flows reveal the general quality of different positions. Sorkin (2018) applied this logic to rank employers and measure compensating differentials.[8] We apply this logic to rank occupations attached to these positions.

In Table 1, we also report two other important features of each occupation that provide useful benchmarks for evaluating our classification. The second column reports the log-deviation of median wages in each occupation from the national median—an objective measure of a "good" occupation.[9] The third column presents a classification based on task intensity. Here, RC represents "routine cognitive,"

[6]Employment shares for certain occupations change dramatically going from December 2002 to January 2003, even when using the IPUM-CPS harmonized variable.

[7]We also study the net inflows for all possible combinations of occupation pairs i and j in Appendix A.

[8]In particular, Sorkin (2018) formally shows how the preferences of workers, which can include idiosyncratic components, can be aggregated into employer-level rankings. Although we do not repeat his logic here, the same argument can be used for ranking occupations.

[9]Wage information comes from the 2006 Occupational Employment and Wage Statistics at the US Bureau of Labor Statistics (BLS).

NRC represents "non-routine cognitive," RM represents "routine manual," and NRM represents "non-routine manual." This type of task-based categorization has been employed in the literature on labor market polarization, such as Acemoglu and Autor (2011). In this table, we follow Carrillo-Tudela and Visschers's (2023) categorization.

Overall, occupations with positive net inflows also tend to have higher median wages and are predominantly NRC. However, there are notable exceptions. For example, the legal occupation has a negative albeit small inflow despite having one of the highest median wages. This phenomenon likely occurs not because it fails to attract people, but because qualification requirements artificially limit inflow into this occupation. Conversely, several occupations have positive net inflows through EE transitions and are thus categorized as attractive despite having median wages below the overall median: "Healthcare support," "Office and Administrative support," "Production," and "Transportation and material moving." All RM occupations are in the attractive-occupation group, while some RC and NRM occupations are categorized as nonattractive. Thus, our flow-based categorization is related to but meaningfully distinct from both wage-based and task-based categorizations. In Appendix F, we perform a robustness analysis under alternative classifications using this additional information for the business cycle features analyzed in the next section.[10]

3 Occupational flows over the business cycle

In this section, we analyze how business cycles affect the mobility of workers across occupations. First, consider workers who experience EE transitions. Suppressing the time subscript to simplify our notation, we first compute the net inflow to A occupations:[11]

$$F_{EE,A} \equiv \frac{EE_{NA} - EE_{AN}}{E_A}, \tag{2}$$

[10]In addition to wage and task intensity classification, we perform an extensive robustness analysis that includes classifications based on certification and licensing requirements, and finer 3-digit SOC classifications.

[11]A occupations are identified from Table 1 with positive average net inflow rates via EE transitions, which uses data after the 2002 changes in the occupational coding scheme.

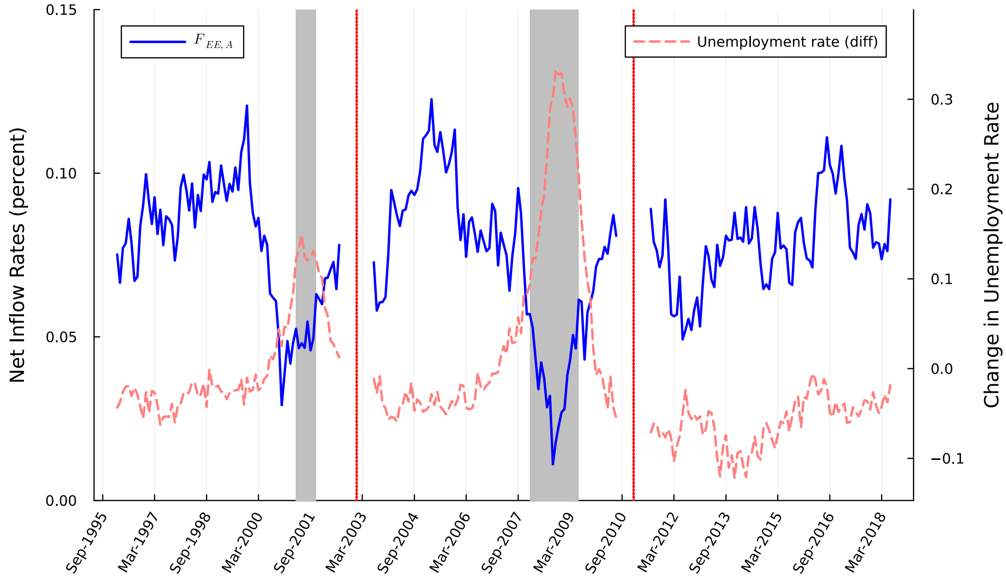

Figure 1: Net inflow rates into attractive occupations via EE transitions. Gray areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

where EE_{ij}, for $i, j = A, N$ and E_A are defined analogously to equation (1). Similarly, for the net inflow to N, we compute

$$F_{EE,N} \equiv \frac{EE_{AN} - EE_{NA}}{E_N}, \tag{3}$$

where the denominator E_N is the size of employment in N occupations. Because we have only two categories, $F_{EE,A}$ and $F_{EE,N}$ always have opposite signs, although their magnitudes can differ due to different denominators.

Figure 1 plots the time series of $F_{EE,A}$. The red vertical lines correspond to the two SOC changes that occurred in 2002 and 2010. These changes divide our series into three subperiods. The monthly series is smoothed with a 12-month moving average within each of these subperiods. To highlight cyclical movements, we plot the changes in the national unemployment rate (measured on the right axis) and shade the NBER recession periods on the same figure. The graph shows two facts: (i) $F_{EE,A}$ is positive everywhere, and its average is 0.07%; and (ii) $F_{EE,A}$ declines during both recessions. The first fact implies, on average, workers climb the *occupation ladder*: they tend to move from nonattractive to attractive occupations. This fact is not surprising given how we constructed attractive versus nonattractive occupations. The second fact, which is our main finding, is less obvious and more important: the occupation ladder is cyclical. In recessions, the net movement from N to A slows down.[12] As we elaborate on below, this fact suggests a novel cost of recession: a recession impedes workers from moving to a better occupation.

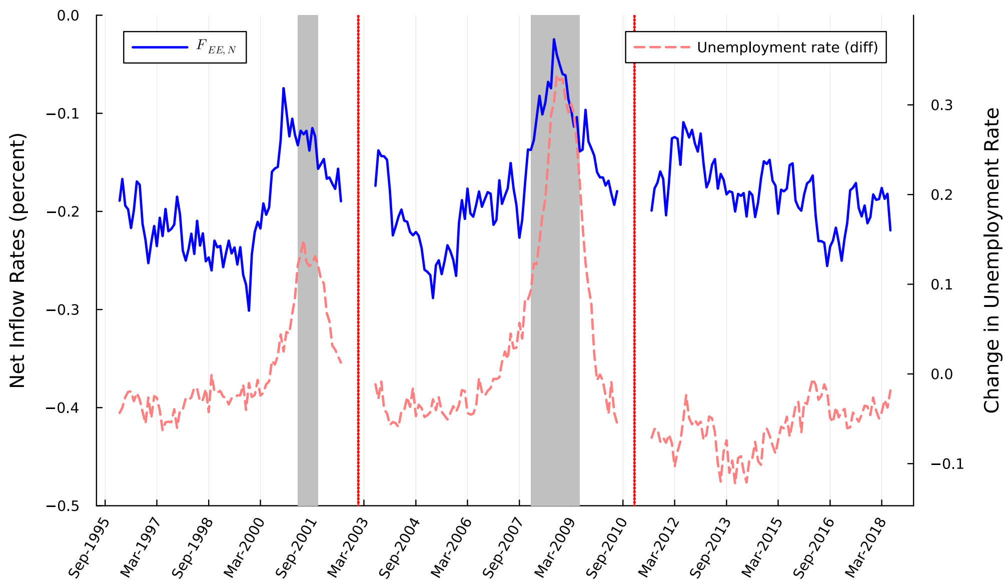

Figure 2: Net inflow rates into nonattractive occupations via EE transitions. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

Figure 2 plots $F_{EE,N}$. As discussed above, by construction, $F_{EE,N}$ has the opposite sign of $F_{EE,A}$. However, it has a larger magnitude in percentage terms, averaging at -0.18%, because the size of employment in N occupations is smaller. The figure shows that the cyclicality is similar (with the opposite sign) to $F_{EE,A}$. The interpretation is similar to the above: during a recession, the net movement from attractive occupations to nonattractive occupations goes up; that is, the ascent of the occupation ladder slows down.

[12] Appendix B demonstrates our finding is robust to looking at the share of the flows instead of the share of stocks. Appendix C constructs similar plots for different gender, age, and education groups. The sharp drop before the 2001 crisis is largely driven by females, and the series for males better aligns with the timing of the 2001 recession. See Figure C.16.

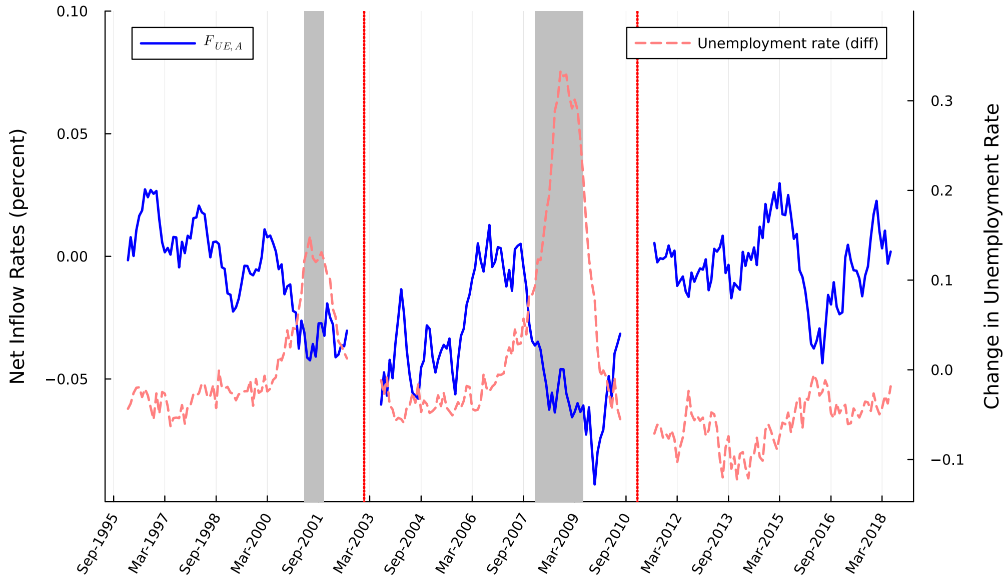

Figure 3: Net inflow rates into attractive occupations via UE transitions. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

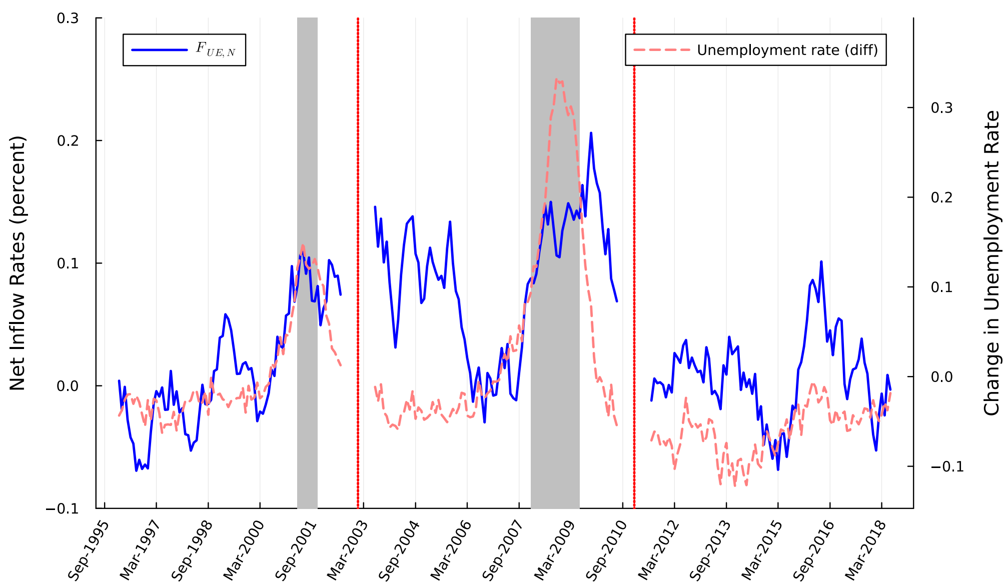

Figure 4: Net inflow rates into nonattractive occupations via UE transitions. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

Figures 3 and 4 plot the same objects as Figures 1 and 2 but with the UE flows. That is, they measure the occupational switch upon job finding from unemployment. Let the total number of unemployed workers who find a job and change occupations be UE_{ij} for $i, j = A, N$. Then, the net inflow rate from nonattractive to attractive occupations via UE transitions is given by

$$F_{UE,A} \equiv \frac{UE_{NA} - UE_{AN}}{E_A}. \tag{4}$$

For the net inflow to N via UE transitions, we compute

$$F_{UE,N} \equiv \frac{UE_{AN} - UE_{NA}}{E_N}. \tag{5}$$

The average values of $F_{UE,A}$ and $F_{UE,N}$ are -0.02% and 0.04%, respectively. Note that the signs are different from the corresponding net inflow rates via EE transitions. This finding implies that unemployed workers, on average, move down the occupation ladder.

The cyclicality in Figure 3 is similar to Figure 1, and the cyclicality of Figure 4 is similar to Figure 2, but the cyclical patterns are weaker with UE than EE.[13] A UE-related occupational switch is less related to the cycle: similarly to Carrillo-Tudela and Visschers (2023), two forces are at work here. On the one hand, recessions lead to fewer openings, reducing opportunities to transition into a more attractive occupation. On the other hand, during recessions, workers tend to be unemployed longer. The longer the duration, the more likely the worker is to take a job in any occupation, including the possibility of moving down the occupation ladder.

4 Separation, vacancy, and other statistics

In this section, we explore additional data features for attractive and nonattractive occupation groups to understand the cyclicality of occupation ladder.

[13] Appendix F.3 illustrates that the cyclicality in the UE-related occupational transition (Figure 3) is largely influenced by the behavior of net inflows into routine-manual occupations.

4.1 Separations by occupation group

First, we construct measures of job separations to unemployment (EU transitions) by occupation category using the monthly CPS data. Specifically, we calculate the fraction of employed individuals in a given occupation category who lose their jobs in the next month and become unemployed.

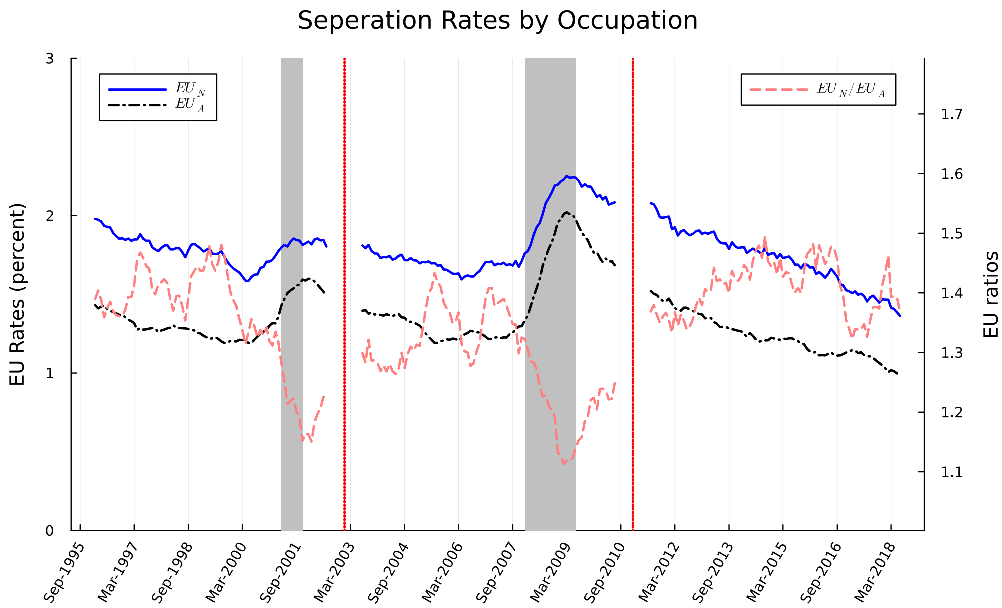

Figure 5: Separation rates into unemployment by attractive and nonattractive occupations. The separation-rate ratio (nonattractive to attractive) is shown on the right scale. Gray shaded areas correspond to NBER recessions, and the vertical red-dotted lines correspond to the changes in SOC. The monthly series is smoothed with a 12-month moving average.

Figure 5 shows the monthly plot of these series for attractive and nonattractive occupations. Employed individuals in attractive occupations are, on average, less likely to lose their jobs and become unemployed. The average monthly separation rate for attractive occupations is about 1.35%, whereas an average of about 1.79% of the employed individuals in nonattractive occupations lose their jobs every month and become unemployed. These numbers indicate that attractive-occupation jobs are more stable than nonattractive-occupation jobs. The mobility pattern suggests that job stability can be one of the benefits of working in an attractive occupation. This result is consistent with Jarosch (2023): workers go through EE transitions to move to more stable jobs.

Transition rates to unemployment are countercyclical for each occupation group. However, the increase in EU rates during recessions is larger for attractive occupations. To highlight this fact, we plot the ratio of separation rates for nonattractive occupations to that of attractive occupations on the right scale in Figure 5. The average value of the ratio is around 1.34, indicating the separation rates to unemployment for nonattractive occupations are 34% more than that of attractive occupations. The ratio falls to about 1.15 during both recessions, which is a significant drop relative to normal times.

In sum, Figure 5 suggests attractive occupations are more stable but also more sensitive to business cycles. This pattern indicates that although attractive occupations can generate more surplus from the job-worker match, the surplus is more cyclical than that of nonattractive occupations. This mechanism is also consistent with the net flow pattern we observed in Section 3.[14]

4.2 Vacancies, unemployment, and market tightness by occupation group

Here, we examine the same mechanism through the lens of vacancies. If the relative match surplus of attractive occupations is smaller during recessions, this movement should also be reflected in labor demand. One popular indicator of labor demand is vacancies.

To explore the behavior of vacancies, we use occupation-level vacancy data from the Conference Board's Help Wanted Online (HWOL) database. The HWOL database provides total vacancies for each of the two-digit occupation categories in the SOC 2010 classification system, which correspond to major occupation groups in the Census classification. The data span a period from May 2004 to October 2018, covering the Great Recession. We supplement this data with the number of employed and

[14]In the context of the Diamond-Mortensen-Pissarides model, a small surplus is often associated with a higher volatility, as was emphasized by Hagedorn and Manovskii (2008). Our results on the comparison between attractive and nonattractive occupations suggest there are different mechanisms at work here. Despite more frequent separation from nonattractive occupations, which suggests a smaller surplus associated with these occupations, these occupations exhibit less cyclical sensitivity. This result highlights the difference in exposure to aggregate shocks across occupations, suggesting that attractive occupations are likely more exposed to these shocks.

unemployed persons in the US by occupation obtained from the BLS database.[15]

In Figure 6, we plot vacancy rates calculated as the ratio of total vacancies in a given occupation group to the labor force size in that occupation group. On average, vacancy rates are higher for attractive occupations, implying higher demand. For both occupation groups, the vacancy rates sharply decline during the Great Recession and then slowly recover. The series also shows a mild upward trend.

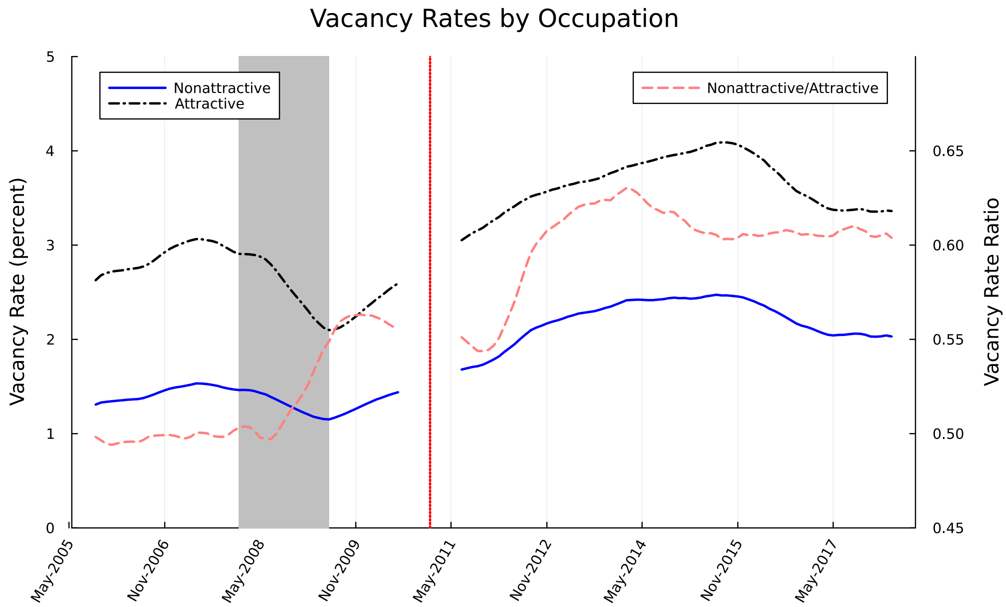

Figure 6: Vacancy rates for attractive and nonattractive occupations. The vacancy-rate ratio (nonattractive to attractive) is shown on the right scale. Gray shaded areas correspond to NBER recessions, and the vertical red-dotted line corresponds to the change in SOC. The monthly series is smoothed with a 12-month moving average.

Although vacancy rates declined during the Great Recession for both occupation groups, the decline is higher for attractive occupations. In Figure 6, the dashed red line, measured on the right scale, shows the ratio of vacancy rates for nonattractive occupations to that of attractive occupations. The ratio moves from 0.50 to about 0.55 during the Great Recession, suggesting that the negative impact of the Great Recession on labor demand is more pronounced for attractive occupations. This movement is consistent with the surplus from attractive occupations being more sensitive to business-cycle conditions. The movement of labor demand, in turn, affects

[15]The original series on the BLS database are not seasonally adjusted. We adjust them using the Census X-11 suite, which the BLS also uses for other time series.

the job- and occupation-switching behavior highlighted in Section 3.

The chances of finding a job depend not only on the number of available vacancies, but also on the number of job seekers. In Figure 7, we plot the unemployment rate for each occupation group calculated using the BLS series. The unemployment rate is countercyclical and is higher among the nonattractive occupations. The unemployment rate increases faster for attractive occupations during the Great Recession. To show this fact more clearly, we plot the ratio of the unemployment rate of the nonattractive occupations to attractive occupations in Figure 7. This ratio, measured on the right scale, shows that the share of unemployed who had an attractive occupation in their previous jobs increased more than nonattractive occupations during the Great Recession. Once again, this result is consistent with the relative decline in match surplus for attractive occupations during recessions.

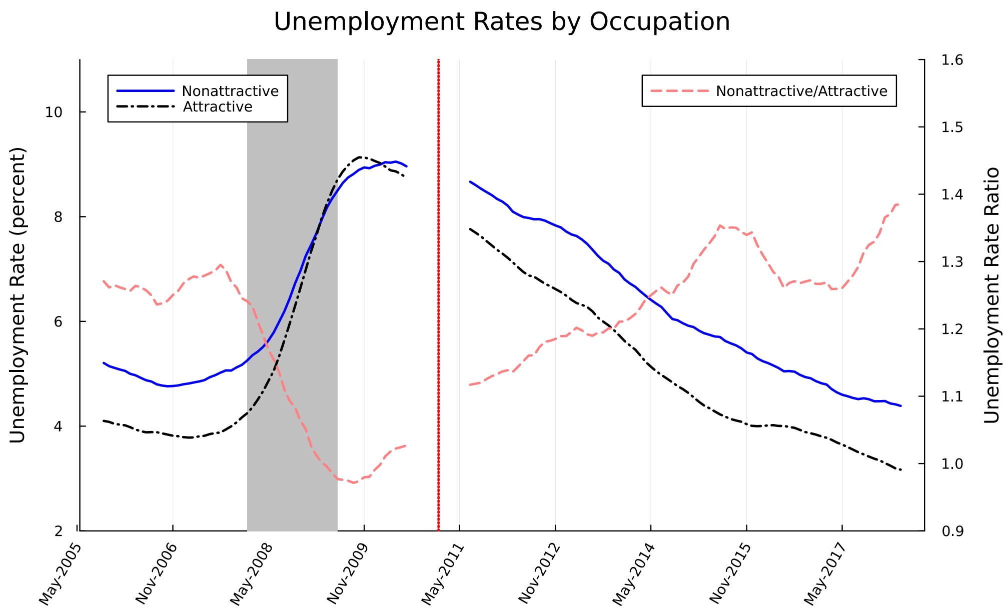

Figure 7: Unemployment rates for attractive and nonattractive occupations. The vacancy-rate ratio (nonattractive to attractive) is shown on the right scale. Gray shaded areas correspond to NBER recessions, and the vertical red-dotted line corresponds to changes in SOC. The monthly series is smoothed with a 12-month moving average.

In a standard matching model, the frictions in the labor market are summarized by an aggregate matching function. Standard assumptions on the matching function imply that a worker's job-finding rate depends positively on the market tightness, defined as the ratio of vacancies to the unemployment rate. Our analyses of vacancy and unemployment rates both suggest that finding an attractive-occupation job becomes relatively more difficult during recessions. Combining our data on vacancies and unemployment, we construct a measure of market tightness separately for attractive and nonattractive occupations. We plot these time series in Figure 8 together with their ratio measured on the right scale. The figure shows that market tightness is procyclical for both occupation groups, but it is cyclically more sensitive for attractive occupations.

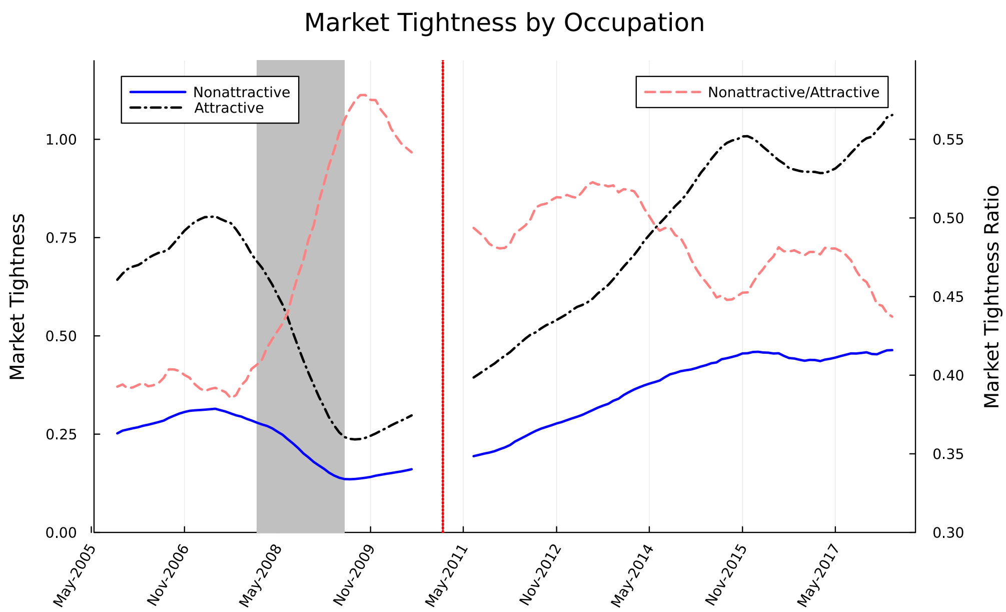

Figure 8: Market tightness for attractive and nonattractive occupations. The market-tightness ratio (nonattractive to attractive) is shown on the right scale. Gray shaded areas correspond to NBER recessions, and the vertical red-dotted line corresponds to changes in SOC. The monthly series is smoothed with a 12-month moving average.

5 Another dataset: NLSY97

In Section 3, we found that the net flow into attractive occupations declines in recessions. In Section 4, other labor market statistics, such as separations and vacancies, point to more cyclical surplus for jobs in attractive occupations. To further supplement our findings with the CPS data, we use data from the National Longitudinal Survey of the Youth (NLSY).

We use the recent 1997 cohort, which is a representative sample of about 9,000 American youth born between 1980 and 1984. This dataset provides detailed employment information for these individuals from the first interview in 1997 to this day. Despite its small sample size and limited age variation, this dataset has at least three advantages that can complement the CPS data for our purposes. First, we note relatively big jumps in the share of some occupation groups in the CPS data after the major change implemented in 2003, even in the harmonized IPUMS-CPS occupation variable. NLSY uses the 2002 Census occupational coding scheme for all the survey years and is not subject to such occupational classification errors. Second, the data contain information about wages, which allows us to relate wage gains upon EE transitions to occupational switch.[16] Finally, the ability to follow an individual's employment history for a long time enables us to explore the occupation ladder over the life cycle of an individual.

In constructing our sample, we exclude job spells that began before the individual turned 18 years old. As the individuals in our sample age over time, their educational attainment increases too. To account for the potential effects of education, we perform our analysis in this section also with a subset of individuals who have never obtained a degree beyond high school in any survey year. This low-education group is particularly helpful because most of these individuals have completed their schooling by the age of 18.

5.1 Results on worker stocks and flows

In Figure 9, the blue curve shows the share of attractive occupations in total employment over time. The series starts in January 1999, when these individuals are between the ages of 15 and 19, and many of them are still attending school.[17] The series ends in December 2018 when these individuals reach their prime ages. To compare our calculations with the entire population, we plot the share of attractive occupations in total employment from our CPS sample for the corresponding month. Note that the

[16] Another alternative we considered is the Survey of Income and Program Participation (SIPP). Although SIPP panels have much larger sample sizes, they unfortunately do not provide a good coverage for recession years. Most importantly, the 2004 SIPP panel ends in January 2008, and the next SIPP panel (2008) starts in September 2008, which excludes the first three quarters of the Great Recession.

[17] Sample sizes significantly drop if we start the series one year earlier.

NLSY time series tracks the life cycle and the calendar time simultaneously, because it tracks a single cohort.

Two patterns emerge from Figure 9. First, individuals climb the occupation ladder over the life cycle, with improvements concentrated early in their careers. Second, the progress among the low-education group is also large and significant. The employment share of attractive occupations in this group increases from about 58% at the start of their careers to 75% when they reach prime ages.

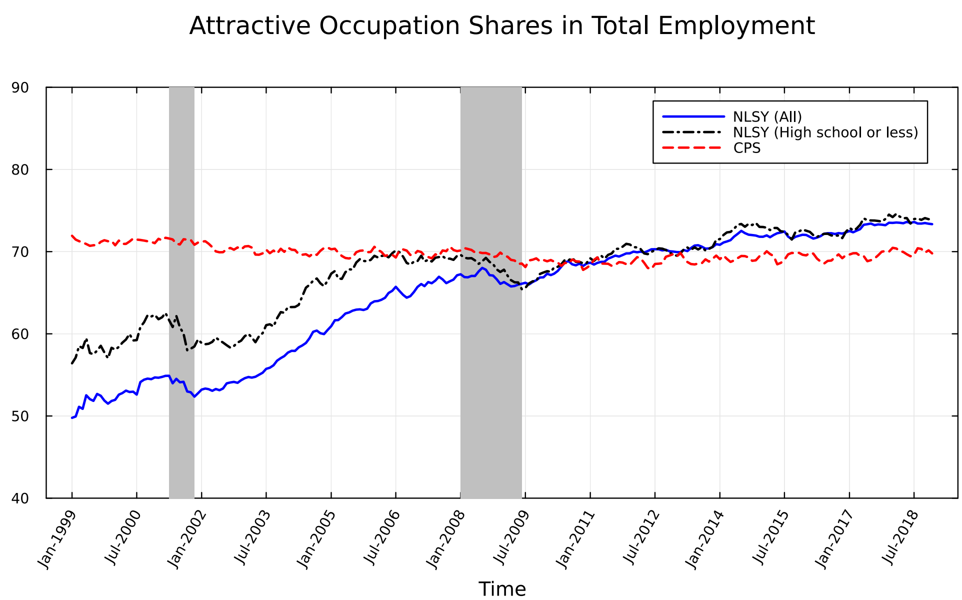

Figure 9: Shares of attractive occupations in NLSY97. Gray areas correspond to NBER recessions. The monthly series is smoothed with a 12-month moving average.

Next, we plot net inflows to attractive occupations via EE transitions in Figure 10. As in the CPS data, the net inflow rates via EE transitions are mostly positive and relatively higher when they are younger. The second observation partly reflects the fact that most of these individuals are employed in a job with an attractive occupation at older ages. More importantly, the net inflow rates drop sharply during both recessions. Therefore, the observations in Section 3 are robust. This pattern is more pronounced for those with a high school degree or less.

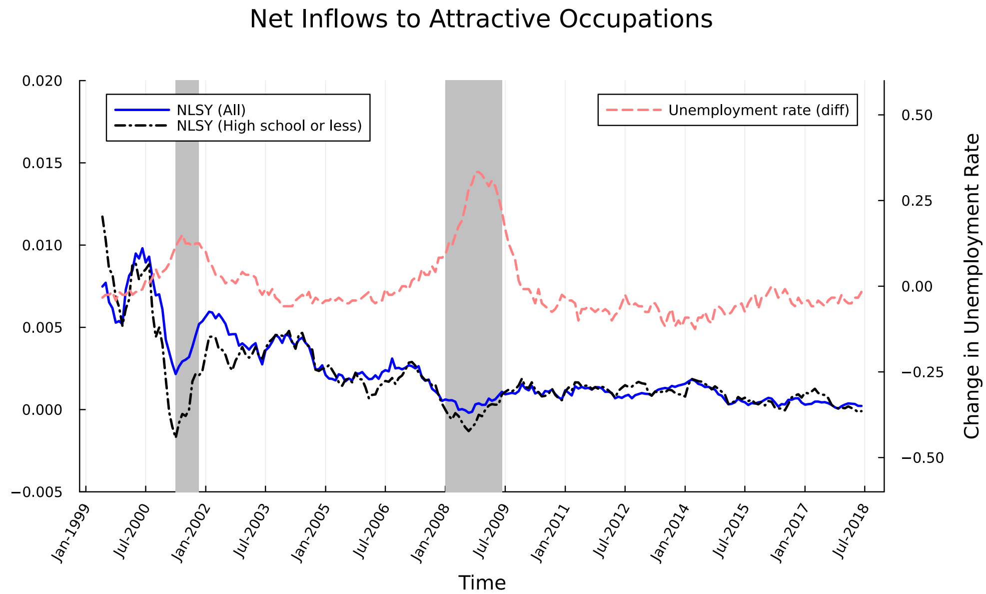

Figure 10: Net inflow rates into attractive occupations via EE transitions in NLSY97. Gray areas correspond to NBER recessions. The monthly series is smoothed with a 12-month moving average.

5.2 Hourly wages

Another advantage of the NLSY data is that the hourly wage rate is available, and we are able to calculate wage gains upon an EE transition. We calculate real hourly wages by dividing the reported nominal hourly wages by the Consumer Price Index. The calculation process for hourly wages produces extremely low and extremely high wage-rate values. Therefore, we drop observations with very high (more than \$400) and very low wage rates (less than \$1). Table 2 provides a cross-tabulation of the log difference between the real hourly wage in the old and new job following an EE transition. In the second column, we calculate these wage gains for individuals with a high school degree or less.[18] Overall, wage gains are sizable in our sample and average 10.5%. Our calculations are larger than some of the previous studies using other data sources. Tjaden and Wellschmied (2014) report that wage gains from EE transitions are on average 3.3% in SIPP data. Using data on earnings and hours from Longitudinal Employer-Household Dynamics (LEHD) data, Hahn et al. (2021)

[18]In Appendix D, we compute the statistics in this Section for higher educational attainment levels.

	All	High school or less
Overall	0.105	0.089
AA	0.116	0.090
NN	0.062	0.073
NA	0.282	0.244
AN	-0.101	-0.111

Table 2: Wage gains at EE transitions: Cross tabulation of the log real hourly wage difference between the old and the new jobs after an EE transition.

report that wage gains from EE transitions averaged 6.2%. These differences partly reflect how we construct our sample. For example, Hahn et al. (2021) also analyze a sample of individuals who entered the labor market in 2010 and track their labor market experience for the next seven years. For this entry cohort, they calculate that the average wage gains upon an EE transition is on average 9.1%.[19] In our sample, the average wage gain via EE transitions in the first seven years (from 1999 to 2005) is 8.8%, which is close to the number calculated in Hahn et al. (2021) for the 2010 entry cohort.

A more striking feature of Table 2 is that these wage gains are much larger when the individuals move from nonattractive to attractive occupations, but it is *negative* when they move from attractive to nonattractive occupations via EE transitions. To further explore whether these wage gains are driven by individual characteristics, we estimate the following regression equation:

$$\Delta w_{it} = \tau_{it}\beta + X_{it}\gamma + \varepsilon_{it}, \tag{6}$$

where Δw_{it} is the log difference between the real hourly wage rate in the new and the old job when individual i changes employers in month t. τ_{it} is a set of indicators for occupational switch and β is the associated vector of coefficients. The set of controls, X_{it}, includes indicators for demographics and job-specific characteristics in the old and in the new jobs, such as part-time and government job status. We also included quadratic terms for the completed tenure in the old job, quadratic terms for the experience of the worker at the time of the EE transition, and the log difference

[19] See their Appendix D for a detailed discussion.

Dependent Variable:	Log Hourly Wage Difference	
Sample:	All	High School or Less
AA	0.0770***	0.0892***
	(0.0111)	(0.0154)
NN	0.0358***	0.0804***
	(0.0120)	(0.0179)
NA	0.2427***	0.2450***
	(0.0141)	(0.0204)
AN	-0.1255***	-0.1057***
	(0.0174)	(0.0258)
Fit statistics		
Observations	19,358	6,737
R^2	0.05432	0.04715
Adjusted R^2	0.05353	0.04517

Clustered (caseid) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3: Wage gains and losses from EE transitions: The dependent variable is the log real hourly wage difference between the new and the old job. Other controls include indicators for demographic characteristics (gender, race, and education), part-time and government job status in both the old and new jobs, quadratic terms for actual experience and tenure in the previous job, and the log difference in the national unemployment rate.

in the national unemployment rate in month $t-1$ and t.

The regression results are presented in Table 3. The first column uses all the EE transitions in our sample, and the second column uses only the individuals with a high school degree or less. In both columns, these results confirm our findings in Table 2, even after including controls. We conclude that wage gains and losses associated with occupational switches are not driven by individual and job characteristics.[20] Rather, the wage changes are driven by the nature of occupational characteristics.

[20]In Appendix, we include interaction of τ_{it} with the highest educational attainment of individual i. The results remain qualitatively unchanged, and we estimate larger gains and losses for higher-education groups.

6 Implications for the cost of recessions

In Section 5.2, we demonstrated that occupational switching can have a significant impact on individual wages. The results there potentially underestimate the gains from climbing the occupational ladder, given that nonpecuniary gains can also exist. Such gains have been emphasized in the literature in the context of job ladders—see, for example, Sorkin (2018).

At the macro level, we have already seen that the speed of climbing the occupation ladder slows during recessions. The natural question is: How much do we lose in recessions due to the aggregate slowdown of climbing up the occupation ladder? In this section, we calibrate a simple model to evaluate the importance of the cyclical occupational mobility quantitatively.

6.1 Model

We model the business cycle as a two-point process of a random variable z_t. We construct the boom (g for "good") and the recession (b for "bad") symmetrically below. The transition probabilities $\pi_{zz'} = \Pr[z_{t+1} = z' | z_t = z]$ for $z, z' = g, b$. Here, prime ($'$) represents the next-period variable.

Occupations $o_t \in \{a, n\}$ are modeled as an idiosyncratic state. The occupation $o_t = a$ represents the "attractive" occupation and $o_t = n$ represents the "nonattractive" occupation. The employment state $\epsilon_t \in \{e, u\}$ is also idiosyncratic. The probability of an employed worker becoming unemployed $\sigma_{oz'}$ depends on z' and the occupation o. The probability of an unemployed worker finding a job $\lambda_{oz'}$ and the probability of EE transition $\gamma_{oz'}$ also depend on z' and o. To map the model to our empirical measurement, we assume that an occupational switch occurs upon either EE or UE transition. $p^{UE}_{oo',z'}$ and $p^{EE}_{oo',z'}$ are the conditional switching probabilities of moving from o to o' when the aggregate state is z'.

The expected lifetime wage is defined as

$$V(\epsilon, o, z) = E\left[\sum_{t=0}^{\infty} \beta^t w(\epsilon_t, o_t, z_t)\right], \tag{7}$$

where $E_0[\cdot]$ is the expectation at time 0, $\beta \in (0,1)$ is the discount factor, and

$w(\epsilon_t, o_t, z_t)$ is the income (wage) of a consumer whose occupation is o_t and the employment state is ϵ_t when the aggregate state is z_t.

The lifetime wage can be written recursively as

$$\begin{aligned} V(u, o, z) = {} & w(u, o, z) \\ & + \beta \sum_{z'} \pi_{zz'} \left(\lambda_{o,z'} \left(\sum_{o' \neq o} p^{UE}_{oo',z'} V(e, o', z') + \left(1 - \sum_{o' \neq o} p^{UE}_{oo',z'}\right) V(e, o, z') \right) \right. \\ & \left. + (1 - \lambda_{o,z'}) V(u, o, z') \right) \end{aligned}$$

for unemployed workers and

$$\begin{aligned} V(e, o, z) = {} & w(e, o, z) \\ & + \beta \sum_{z'} \pi_{zz'} \left(\gamma_{o,z'} \left(\sum_{o' \neq o} p^{EE}_{oo',z'} V(e, o', z') + \left(1 - \sum_{o' \neq o} p^{EE}_{oo',z'}\right) V(e, o, z') \right) \right. \\ & \left. + \sigma_{o,z'} V(u, o, z') + (1 - \gamma_{o,z'} - \sigma_{o,z'}) V(e, o, z') \right) \end{aligned}$$

for employed workers. The future values for unemployed workers reflect (i) finding a job and moving to another occupation, (ii) finding a job and staying in the same occupation, and (iii) staying unemployed. The future values for employed workers reflect (i) switching to a new job and changing occupations, (ii) switching to a new job and staying in the same occupation, (iii) losing the job, and (iv) staying in the same job.[21]

6.2 Calibration

One period is one month. The transition probabilities for the aggregate state, $\pi_{zz'}$, is set as

$$\begin{bmatrix} \pi_{gg} & \pi_{gb} \\ \pi_{bg} & \pi_{bb} \end{bmatrix} = \begin{bmatrix} 59/60 & 1/60 \\ 1/60 & 59/60 \end{bmatrix},$$

[21] We do not take a stand on why workers decide to change occupations, in particular when they move to N occupation jobs knowing they pay low. One interpretation is that employed workers face relocation shocks, as emphasized in Jolivet et al. (2006), and these shocks entail moving down the occupation ladder. Carrillo-Tudela et al. (2022) emphasize the role of idiosyncratic shocks to occupational productivity to understand occupational switch.

	$z'=g$	$z'=b$
$\lambda_{az'}$	0.2958	0.2905
$\lambda_{nz'}$	0.3080	0.3069
$\gamma_{az'}$	0.0200	0.0191
$\gamma_{nz'}$	0.0280	0.0268
$\sigma_{az'}$	0.0137	0.0167
$\sigma_{nz'}$	0.0183	0.0196
$p^{EE}_{an,z'}$	0.1556	0.1576
$p^{EE}_{na,z'}$	0.3540	0.3278
$p^{UE}_{an,z'}$	0.2084	0.2043
$p^{UE}_{na,z'}$	0.3518	0.3342

Table 4: Transition probabilities

$w(e,a,g)$	1.177
$w(e,a,b)$	1.177
$w(e,n,g)$	1.000
$w(e,n,b)$	1.000
$w(u,a,g)$	0.400
$w(u,a,b)$	0.400
$w(u,n,g)$	0.400
$w(u,n,b)$	0.400

Table 5: Wages

which approximately reflects the length of one cycle according to the NBER dating. In recent years, we have experienced one cycle every 10 years, and (with the symmetry assumption) the average duration of each state is set at 60 months.

The job-finding probability for the workers in an attractive occupation, $\lambda_{az'}$, is computed as follows. First, we define the boom following the NBER dating. Then, we compute the job-finding probability for f^a_t from the data and decompose f^a_t into the HP trend and its deviation:

$$f^a_t = \tilde{f}^a_t + \hat{f}^a_t,$$

where $\tilde{f}_t^a$ is the HP trend and $\hat{f}_t^a$ is the deviation. Let the sample mean

$$\bar{f}_t^a \equiv \frac{1}{T}\sum_{t=1}^{T} f_t^a.$$

Then, compute λ_{ag} by

$$\lambda_{ag} = \bar{f}_t^a + \frac{1}{N_g}\sum_{t\in G} \hat{f}_t^a,$$

where N_g is the number of periods of NBER booms and G is the set of boom years. Similarly, λ_{ab} can be computed as

$$\lambda_{ab} = \bar{f}_t^a + \frac{1}{N_b}\sum_{t\in B} \hat{f}_t^a,$$

where N_b is the number of periods of NBER recessions and B is the set of recession years. Other probabilities, $\gamma_{oz'}$, $\sigma_{oz'}$, and $p^{EE}_{oo',z'}$ are computed analogously.[22] The results are in Table 4.

Table 4 has some notable outcomes. First, as is well known, the job-finding probabilities for unemployed workers are procyclical, and the separation probability is countercyclical. The EE transition rate is procyclical. Second, as we emphasized in Section 4, attractive occupations are more sensitive to business cycles in that their transition rates change more over the business cycle. Third, note that the probability of an occupational switch is the combination of the job switching (or job finding) and the conditional probability. Both are cyclical. One notable feature of the conditional probabilities is that $p^{EE}_{na,z'}$ and $p^{UE}_{na,z'}$ are strongly procylical. This feature drives the strong cyclicality of the net occupational flow, which we observed in Section 3. Note that our model is based on gross flows and we accordingly target gross flow rates. Net inflow rates are implied by these gross flow rates in equilibrium.

Table 5 summarizes our assumption on wages. In the baseline case, we assume an attractive occupation provides 17.7% better wages than a nonattractive occupation. In Table 2, the wage gain from the NA transition is about 28.2%, and the overall wage

[22]One potential concern is whether the cyclical patterns of transitions are affected by other occupational characteristics. For example, occupations with lower entry barriers may have different cyclical patterns of transitions from high entry barrier occupations. Appendix F.4 examines this particular concern, and we find that the cyclical patterns of net inflows into and out of low-entry barrier occupations look similar to the patterns of aggregate net inflows.

gain is 10.5%. Thus, we estimate the pure effect of moving from N occupation to A occupation as $28.2 - 10.5 = 17.7\%$. We do not incorporate further wage dynamics in the model, so that we can focus on the cost and benefit of occupation switching.[23]

The value of unemployment is 40.0% of the nonattractive occupation wage. The 40% replacement ratio is standard in the literature (Shimer, 2005). This calibration is typically interpreted as the level of unemployment insurance benefit in the US, but can potentially include home production and the value of leisure.

6.3 Results

Let the lifetime wages $V(\epsilon, o, z)$ be computed as in (7). We first consider the counterfactual experiment where all recessions are eliminated by setting all transition probabilities of the idiosyncratic shock to the values during booms. That is, the new probabilities are $\hat{\lambda}_{ab} = \lambda_{ag}$, $\hat{\lambda}_{nb} = \lambda_{ng}$, $\hat{\gamma}_{ab} = \gamma_{ag}$, $\hat{\gamma}_{nb} = \gamma_{ng}$, $\hat{\sigma}_{ab} = \sigma_{ag}$, $\hat{\sigma}_{nb} = \sigma_{ng}$, $\hat{p}^{EE}_{an,b} = p^{EE}_{an,g}$, $\hat{p}^{EE}_{na,b} = p^{EE}_{na,g}$, $\hat{p}^{UE}_{an,b} = p^{UE}_{an,g}$, and $\hat{p}^{UE}_{na,b} = p^{UE}_{na,g}$. Note that our exercise differs from Lucas (1987), who imagines a world without business cycles as one where all macroeconomic variables are replaced by their mean values. Instead, our measurement is the cost of recession—we replace all "recessions" by "booms." In this sense, our calculation follows the tradition of the "plucking view" of Friedman (1964) (also see Dupraz et al., 2024).

Let the counterfactual lifetime wage be $\hat{V}(\epsilon, o, z)$ and define the cost of recession for an individual with state (ϵ, o, z), $\hat{\Delta}(\epsilon, o, z)$, as

$$\hat{\Delta}(\epsilon, o, z) \equiv \frac{\hat{V}(\epsilon, o, z) - V(\epsilon, o, z)}{\hat{V}(\epsilon, o, z)}.$$

This cost of recession includes various factors that make recessions worse compared to booms. The cost here includes a greater loss of income due to more frequent and prolonged unemployment.

To isolate the effect of the change in the movement along the occupation lad-

[23] One potential extension is to include the wage growth while the worker stays in the same occupation due to occupation-specific human capital accumulation. This extension can make a difference if, for example, attractive and nonattractive occupations differ in terms of the speed of human capital accumulation. Here, we abstract from such consideration to parsimoniously model the occupation ladder, and we leave such extension to future research.

der during the recessions, let us run another counterfactual experiment. Here, we maintain the assumption of the first counterfactual experiment for most of the probabilities; that is, the values are set at the boom value. The only exceptions are the conditional occupation switching probabilities $p^{EE}_{oo',z'}$ and $p^{UE}_{oo',z'}$. For $p^{EE}_{oo',z'}$ and $p^{UE}_{oo',z'}$, instead of setting them at the boom value (i.e., the value with $z' = g$), we maintain the probabilistic structure of the original baseline.

Specifically, we set these conditional probabilities so that the unconditional probability of "moving up" or "moving down" the occupation ladder becomes the same as the baseline. Thus, the new conditional probabilities $\tilde{p}^{EE}_{na,b}$ $\tilde{p}^{UE}_{na,b}$, $\tilde{p}^{EE}_{an,b}$, and $\tilde{p}^{UE}_{an,b}$ now satisfy

$$\gamma_{ng}\tilde{p}^{EE}_{na,b} = \gamma_{nb}p^{EE}_{na,b}, \tag{8}$$

$$\lambda_{ng}\tilde{p}^{UE}_{na,b} = \lambda_{nb}p^{UE}_{na,b},$$

$$\gamma_{ng}\tilde{p}^{EE}_{an,b} = \gamma_{nb}p^{EE}_{an,b},$$

and

$$\lambda_{ng}\tilde{p}^{UE}_{an,b} = \lambda_{nb}p^{UE}_{an,b}.$$

In the first equation above, the right-hand side, $\gamma_{nb}p^{EE}_{na,b}$, is the baseline unconditional probability of an occupational switch from the nonattractive occupation to the attractive occupation when the economy is in recession. We set the new conditional probability $\tilde{p}^{EE}_{na,b}$ in the new counterfactual so that the unconditional probability of an occupational switch is the same as in the baseline. Because the EE transition probability in this counterfactual is set at the boom level γ_{ng}, the unconditional probability is as in the left-hand side of (8): $\gamma_{ng}\tilde{p}^{EE}_{na,b}$. Thus, we use the formula in equation (8) for setting the new conditional probability $\tilde{p}^{EE}_{na,b}$. The other three equations are analogous.

The other probabilities are the same as those of the first counterfactual experiment. We define the new welfare cost as

$$\tilde{\Delta}(\epsilon, o, z) \equiv \frac{\tilde{V}(\epsilon, o, z) - V(\epsilon, o, z)}{\tilde{V}(\epsilon, o, z)}.$$

This value measures the cost of recessions *except for* the effect of the occupational ladder. Therefore, the gap between $\hat{\Delta}(\epsilon, o, z)$ and $\tilde{\Delta}(\epsilon, o, z)$ measures the cost of the

	$\hat{\Delta}(\epsilon, o, z)$	$\tilde{\Delta}(\epsilon, o, z)$	$R(\epsilon, o, z)$
(e, a, g)	0.0039	0.0034	12.6
(e, n, g)	0.0040	0.0029	27.6
(u, a, g)	0.0040	0.0034	15.3
(u, n, g)	0.0040	0.0032	23.2
aggregate g	0.0039	0.0033	16.8
(e, a, b)	0.0051	0.0046	8.9
(e, n, b)	0.0052	0.0033	38.0
(u, a, b)	0.0051	0.0046	10.0
(u, n, b)	0.0056	0.0036	35.3
aggregate b	0.0051	0.0042	17.6

Table 6: Welfare costs

changed mobility of the occupation ladder during the recessions. We compute the relative contribution of the occupation ladder (in %) as

$$R(\epsilon, o, z) \equiv \left(1 - \frac{\tilde{\Delta}(\epsilon, o, z)}{\hat{\Delta}(\epsilon, o, z)}\right) \times 100.$$

Table 6 exhibits $\hat{\Delta}(\epsilon, o, z)$, $\tilde{\Delta}(\epsilon, o, z)$, and $R(\epsilon, o, z)$ for the consumer with the individual state (ϵ, o) when the welfare is compared at the aggregate state z. The lifetime wage loss from recessions $\hat{\Delta}(\epsilon, o, z)$ is about 0.3% to 0.6% of income. A significant part of it, about 10% to 40%, is due to the changed speed of movement along the occupation ladder.

The rows with "aggregate" are the average welfare change weighted by the steady-state population in each aggregate state. On average, about 17% to 18% of the cost of recession is due to the occupation ladder. Note, again, our estimate here is a conservative one, because the benefit of moving to a better occupation is likely to be larger than the measured wage gains. Therefore, our experiment demonstrates that slowing down the occupation ladder during a recession is a nonnegligible part of the aggregate cost of recessions.

7 Conclusion

We introduced a new concept, "occupation ladder," in this paper. As in the case of the job ladder, workers tend to move to a better occupation over time. To examine the worker movement along the ladder, we first empirically identified "attractive" occupations and "nonattractive" occupations using the flow movements of workers.

Then, we examined the cyclical behavior of worker flows along the occupational ladder. We found the net flow into attractive occupations is procyclical. That is, workers move up the occupation ladder faster during booms. This fact provides a novel perspective on the cost of recessions.

Other facts on separation and vacancies corroborate the view that the value of an attractive-occupation job is more cyclical than that of a nonattractive-occupation job. For the workers, therefore, the opportunity for a good occupation is hindered by the lack of opportunities during recessions.

Finally, we constructed a simple accounting model to measure the cost of recessions on the workers' lifetime earnings. The overall cost of recessions is about 0.3% to 0.6% of income. In aggregate, the lack of opportunities for climbing the occupation ladder accounts for about 17% to 18% of the overall cost. Our results from NLSY97 indicate that the occupation ladder matters more for young workers. It is an important future research topic to investigate further the scarring effect of recessions through the occupation ladder for young workers.

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Online Appendix for "Occupation Ladders over the Business Cycle"

A Constructing the net inflow rate

When constructing attractive- and nonattractive-occupation groups, we use the information on the net inflow rates via EE transitions in the cross section. In this Appendix, we provide additional details on how we construct these measures and describe their relationship before classifying them as attractive or nonattractive.

The current occupation classification system, SOC 2010, was adopted in 2010 and is based mainly on the 2002 classification system, SOC 2002. The changes in the new classification system are minor, and we do not observe any large difference between the employment shares in December 2009 and January 2010 for a given occupation. However, major changes were implemented in 2002 in the occupation classification relative to the previous one adopted in 1990. Although the IPUMS-CPS database provides a crosswalk for the occupations reported based on the previous classification system, we observe big jumps in employment shares for certain occupations going from December 2002 to January 2003. To avoid potential measurement issues related to changes in the occupations classification system in 2002, we restrict our sample to the basic CPS monthly files that span the period from January 2003 to December 2018. Then, we pool all the observations and calculate the net inflows for all possible combinations of occupation pairs i and j. We perform these calculations separately for UE and EE transitions. Similar to equation 1, we calculate the net inflow rate into occupation i from occupation j as follows:

$$F_{EE,ij} \equiv \frac{EE_{ji} - EE_{ij}}{E_i} \qquad \text{and} \qquad F_{UE,ij} \equiv \frac{UE_{ji} - UE_{ij}}{E_i}.$$

These calculations provide a matrix of net inflow rates for each occupation pair i and j. We represent these rates for EE and UE transitions as grayscale heatmaps in Figure A.11. The rows correspond to the occupation in the current job for EE transitions, and to the occupation in the previous job for UE transitions. The columns correspond to the occupation in the next job. Positive and negative values are distinguished by the upper and lower triangles, respectively. As the absolute value decreases, the color becomes lighter. The diagonal elements are equal to zero by construction and are left blank. The last row displays the overall inflow rate into the occupations in the columns.[24] These overall inflow rates are reported in the first column of Table 1 in the main text.

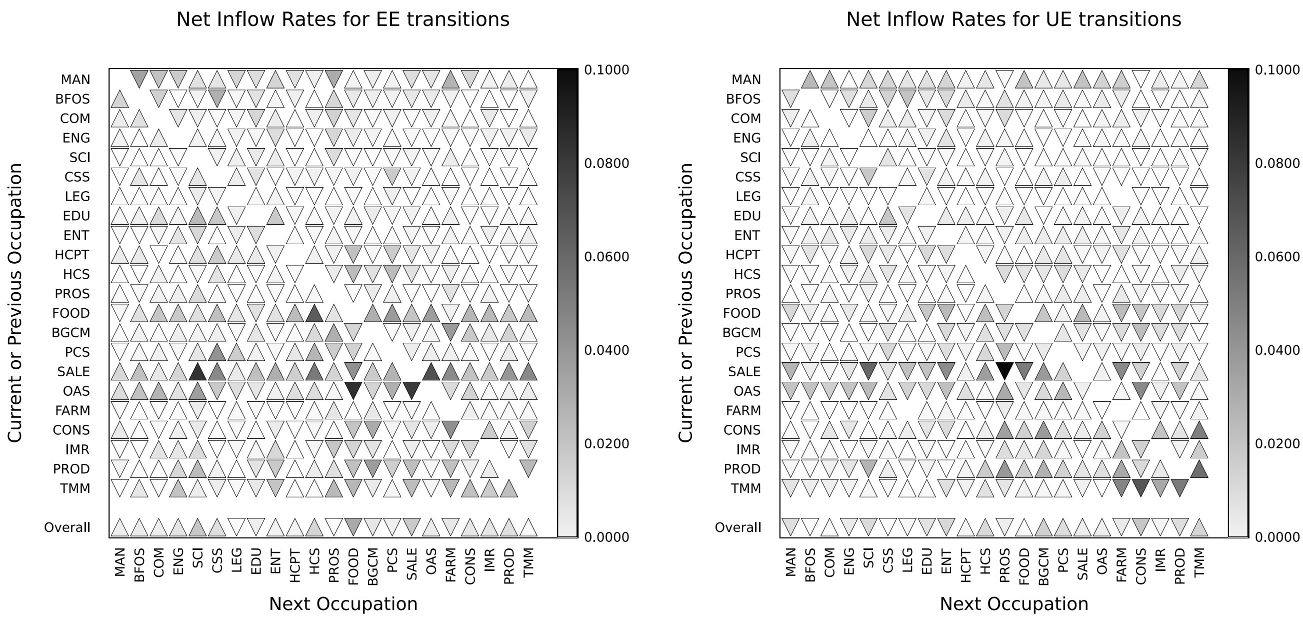

Figure A.11: Net inflow rates via EE and UE transitions by occupation. The occupation categories are based on SOC 2010. The calculations use CPS data from January 2003 to December 2018.

Inspection of Figure A.11 for EE transitions reveals that food preparation and serving ("FOOD") and sales-related ("SALE") occupations lose on net to most other occupation categories, as evidenced by the predominance of lower triangles in their respective columns. However, for UE transitions, these occupation categories gain from many other occupation groups. The opposite pattern occurs for managerial occupations ("MAN"), shown in the first column. The majority of cells are upper triangles in the first column for EE transitions, but almost all cells are lower triangles for UE transitions, implying people move out of managerial occupations when they become unemployed. These contrasting patterns for EE and UE transitions illustrate the importance of studying occupational switches by transition type.

[24] We divide the overall inflow rate by 10 to avoid having a dominant dark color in the last row.

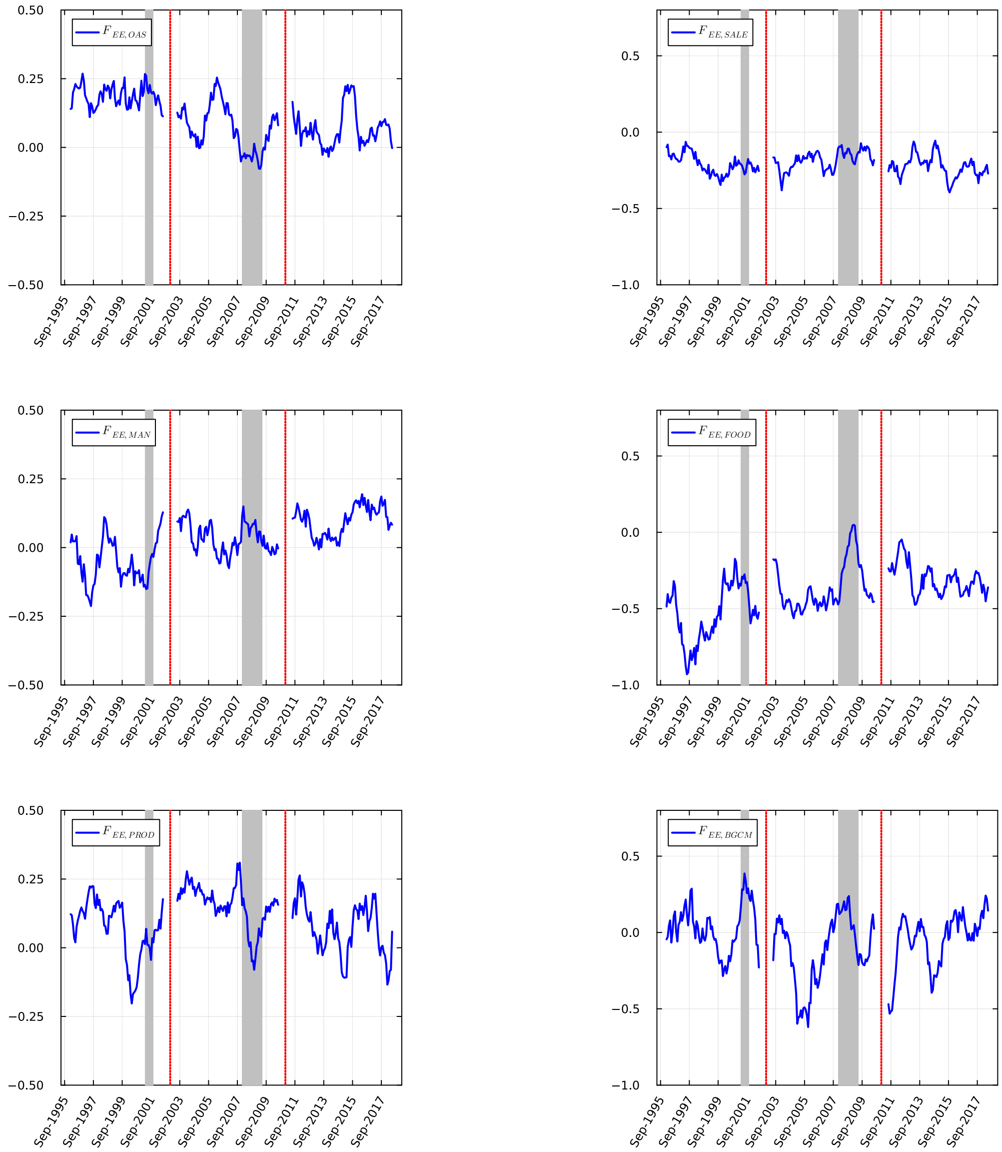

Figure A.12: Net inflows by occupation via EE transitions. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The monthly series is smoothed with a 12-month moving average.

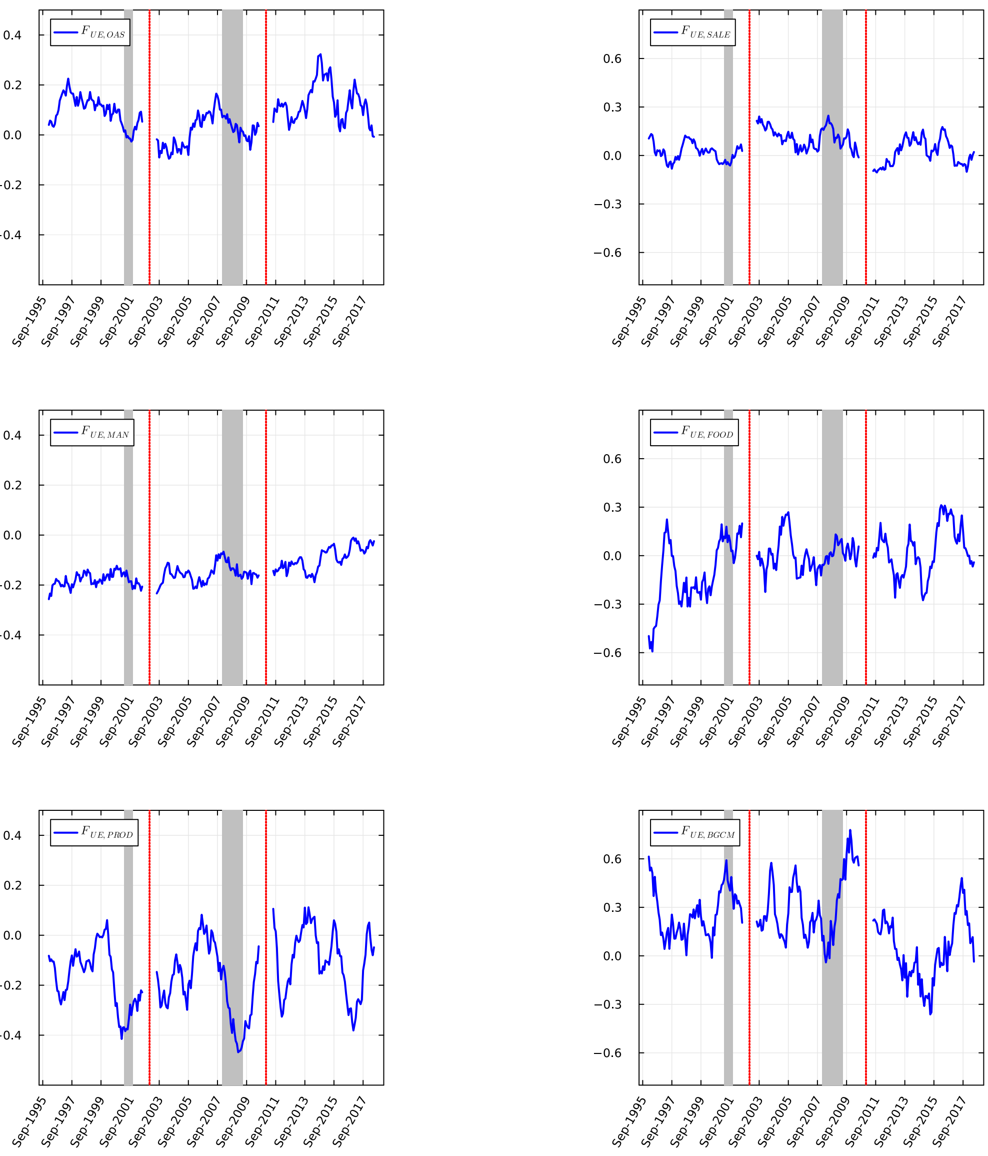

Figure A.13: Net inflows by occupation via UE transitions. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The monthly series is smoothed with a 12-month moving average.

In Figures A.12 and A.13, we plot net inflow rates for selected occupations through EE and UE transitions, respectively. The first column in each figure displays inflow rates for attractive occupations with the largest employment shares, while the second column presents those for nonattractive occupations. Due to sample size considerations, we focus on the three largest occupations in each category. We observe a positive trend in net inflow rates via EE transitions to managerial occupations ("MAN") and a declining trend for office and administrative support ("OAS") positions. Net inflows to managerial occupations change from negative to positive over time.

B Flows as a share of total EE transitions

Here, we calculate net inflows into attractive occupations as a share of total EE transitions. That is, we divide the difference between EE_{NA} and EE_{AN} by total EE transitions:

$$\tilde{F}_{EE,A} = \frac{EE_{NA} - EE_{AN}}{EE}. \tag{9}$$

By construction, $\tilde{F}_{EE,N} = -\tilde{F}_{EE,A}$. Figure B.14 shows net inflows into A occupations as a share of total EE transitions also markedly drop during both recessions. Note that the overall EE rate slows down during recessions. Figure B.14 further shows that among the individuals who make an EE transition during a recession, a smaller fraction is able to climb the occupation ladder.

In Figure B.15, we plot the net inflow rates into attractive occupations as a share of total UE transitions. The cyclicality of this series is similar to the series in Figure 3 in that unemployed individuals tend to move down the occupation ladder. Noting that job-finding rates are lower during recessions, Figure B.15 further implies that, among the unemployed who were able to find a job, proportionally more people are moving down the occupation ladder.

C Flows by demographics

In this section, we plot the net inflow rates from nonattractive to attractive occupations by individual characteristics. We focus on EE transitions. In each figure, all the series correspond to the simple moving average of the original series over a

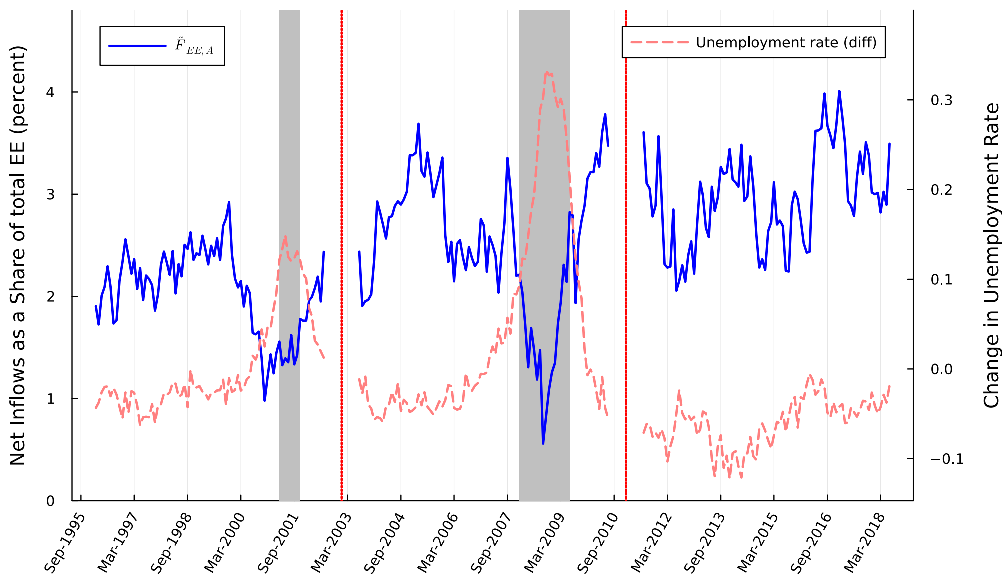

Figure B.14: Net inflows to attractive occupations as a share of total EE transitions. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

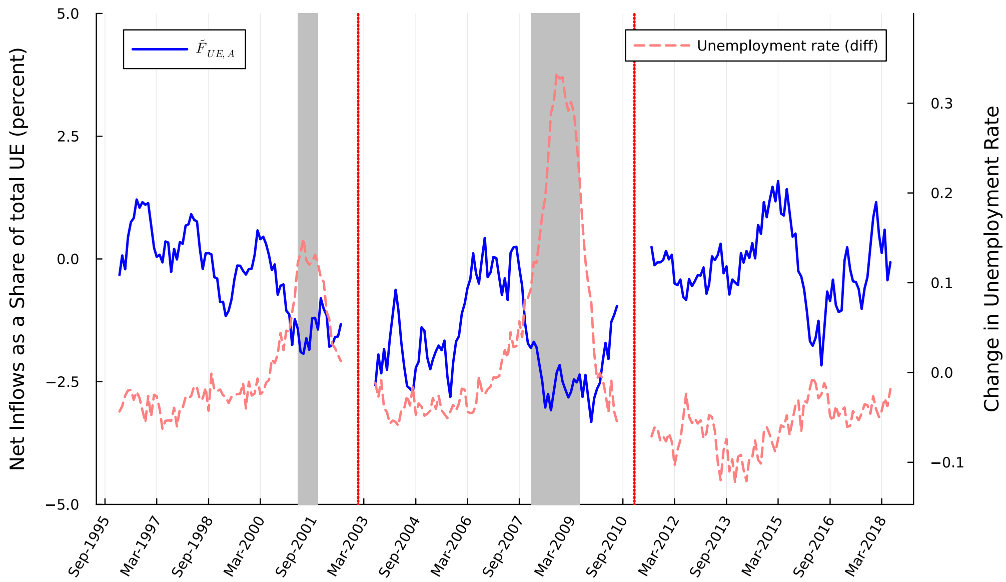

Figure B.15: Net inflows to attractive occupations as a share of total UE transitions. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

12-month period.

Figure C.16 shows the breakdown of net inflow rates into attractive occupations via EE transitions by gender. The general pattern in net inflow rates is present for both men and women, although women are more likely to move to an attractive occupation when they switch jobs, and their career progress along the occupation ladder is more sensitive to business cycles.

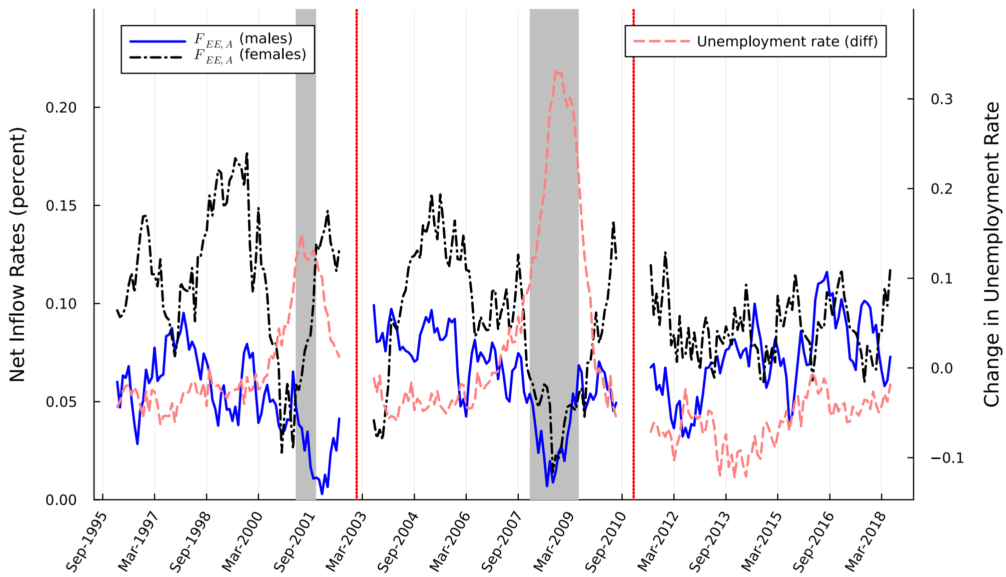

Figure C.16: Net inflow rate into attractive occupations via EE transitions by gender. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

Figure C.17 shows the breakdown of net inflow rates into attractive occupations via EE transitions by different age groups. Young individuals (aged between 16 and 24) are a lot more likely than older individuals to move from a nonattractive occupation to an attractive one. This finding accords with our analysis of NLSY data displayed in Figure (10). The net inflow rate is on average about 0.4%, which is roughly eight times larger than that of an individual aged between 25 and 39. The net inflow rates for young individuals are also more sensitive to the business cycles. For example, at the peak of the Great Recession, the inflow rate from nonattractive occupations to attractive occupations was slightly below 0.2% for young individuals, compared with about 0.4% at the start of the recession. Given that the early career prospects play an important role in one's future labor market experience, Figure C.17 highlights the importance of a cyclical occupation ladder.

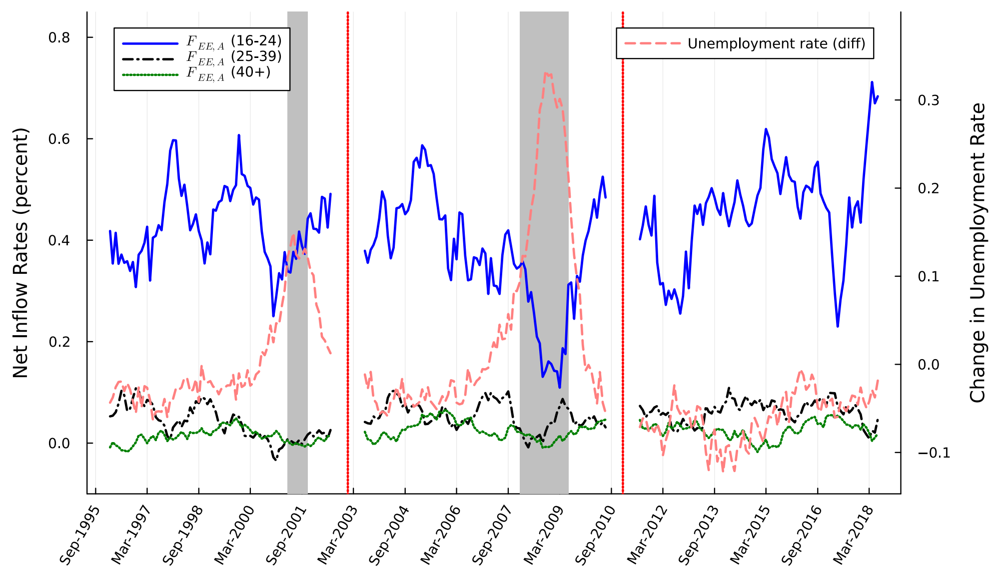

Figure C.17: Net inflow rate into attractive occupations via EE transitions by age groups. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

Figure C.18 shows the net inflow rates into attractive occupations via EE transitions for different education groups. We define three education groups: (i) high school or less, (ii) some college education, and (iii) college and above. On average, people at every education level tend to move from nonattractive to attractive occupations, and an inverse relationship exists between the education levels and the inflow rates to attractive occupations. This inverse relationship is possibly due to the fact that highly educated individuals are already in attractive occupations. Nonetheless, all education categories are negatively affected by the business cycles in that net inflow rates from nonattractive to attractive occupations decline during recessions.

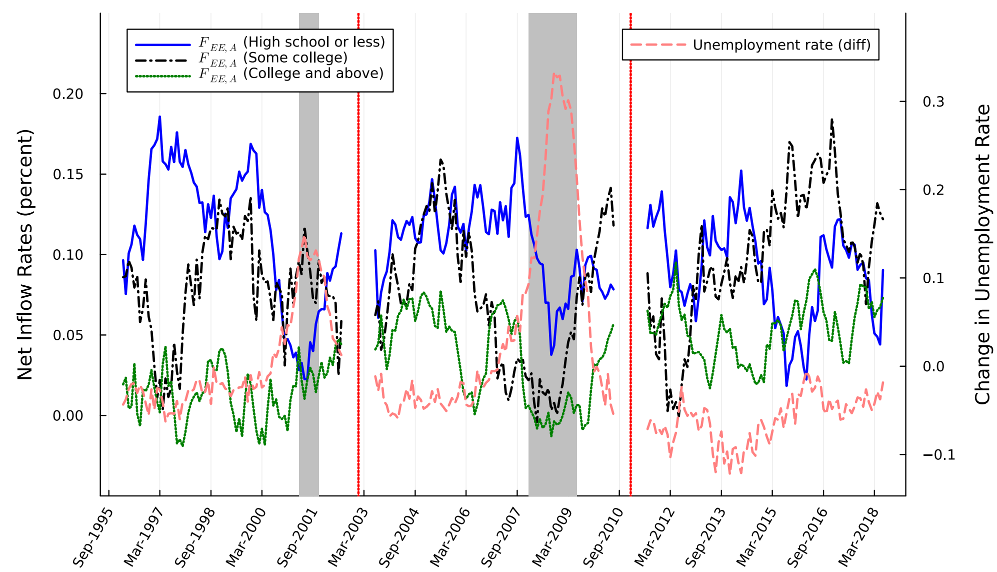

Figure C.18: Net inflow rate into attractive occupations via EE transitions by education: less than high school (HS), high school, some college (COL) education, and college degree and above. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

	High school or less	Some college	College and above
Overall	0.088	0.113	0.155
AA	0.089	0.114	0.162
NN	0.071	0.071	0.113
NA	0.243	0.312	0.320
AN	−0.111	−0.100	−0.018

Table 7: Wage gains at EE transitions by education: Cross tabulation of the log real hourly wage difference between the old and the new job after an EE transition.

D Wage gains by educational attainment

In this section, we divide our NLSY sample by different education groups defined as in the previous section. As before, we drop the job spells that start before the individual turns 18 years old. In addition, we drop the job spells when they complete their highest degree to mitigate the effects associated with schooling decisions.

In Table 7, we report the sample averages of wage gains by each education group.[25] The qualitative results remain unchanged, although we observe larger gains but smaller losses for higher education groups.

To further explore wage gains by education, we estimate the following regression equation:

$$\Delta w_{it} = \sum_{educ} educ_i \tau_{it} \beta^{educ} + X_{it}\gamma + \varepsilon_{it}, \tag{10}$$

where $educ$ denotes the three education categories defined above. $educ_i$ is a dummy variable that is equal to 1 if the highest educational attainment of individual i is equal to the given education category, and β^{educ} is the associated vector of education-specific coefficients. The set of controls is as in Table 3.

The regression results are presented in Table 8. The results are similar to our findings in Table 3. In particular, individuals experience wage losses when they move from an attractive occupation to a nonattractive occupation in every education category, whereas they gain when they move in the opposite direction or remain in

[25] Note that a small number of individuals graduated from high school and achieved their highest degree ever after reaching age 18. By construction, the job spells after age 18 but before graduating from high school are excluded from our sample. Therefore, the wage gains for the education group with a high school degree or less are slightly different than those reported in Table 2.

the same occupation. Moreover, these gains and losses increase with educational attainment, with the exception of NN transitions.

E Gross Flows

In this section, we provide details about the gross flows that underlie the net flows studied in the main text. We focus on gross flows into and out of attractive occupations, since the analysis for nonattractive occupations follows a similar pattern, with only the denominator changing in the calculations. Similar to equations (2) and (3), we define $F_{EE,A+} = EE_{NA}/E_A$ and $F_{EE,A-} = EE_{AN}/E_A$ as the gross inflow and outflow rates for attractive occupations via EE transitions, respectively. Figure E.19 shows the time series of these flow rates. Both gross flows decline during recessions, consistent with the procyclical nature of job-to-job transitions. However, outflows from attractive occupations decline more rapidly than inflows, generating procyclical net inflows into attractive occupations.

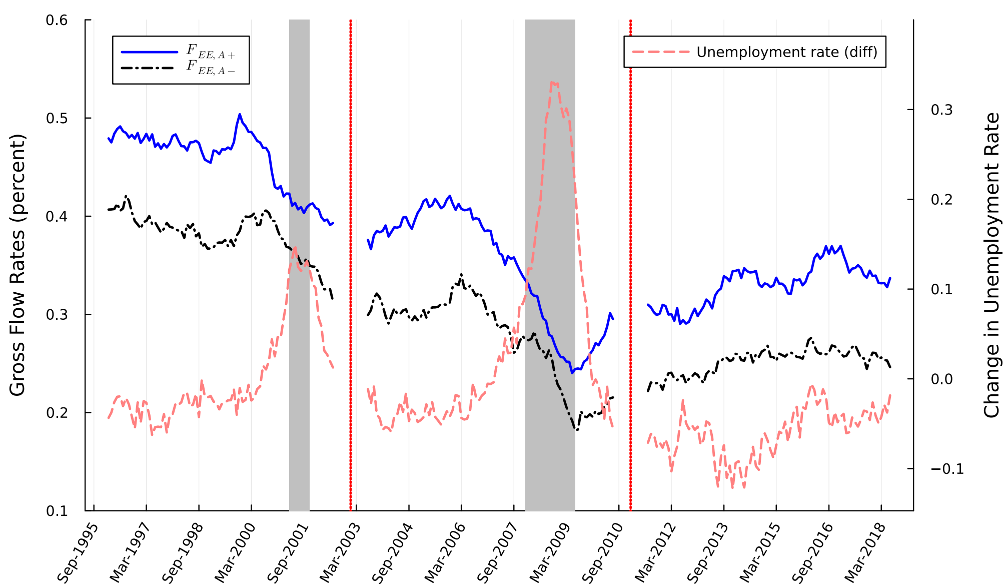

Figure E.19: Gross flow rates into and out of attractive occupations via EE transitions. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

Dependent Variable:	Log Hourly Wage Difference
High school or less:	
AA	0.1017***
	(0.0122)
NN	0.0943***
	(0.0173)
NA	0.2556***
	(0.0190)
AN	-0.0892***
	(0.0246)
Some college:	
AA	0.1282***
	(0.0164)
NN	0.0915***
	(0.0242)
NA	0.3234***
	(0.0398)
AN	-0.0791**
	(0.0396)
College and above:	
AA	0.1706***
	(0.0169)
NN	0.1303***
	(0.0296)
NA	0.3248***
	(0.0321)
AN	0.0034
	(0.0405)
Observations	12,877
R^2	0.04668
Adjusted R^2	0.04505

Clustered (individual) standard-errors in parentheses.
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 8: Wage gains from EE transitions by education: The dependent variable is the log real hourly wage difference between the old and the new job. Other controls include indicators for demographic characteristics (gender and race), part-time and government job status in both the old and new jobs, quadratic terms for actual experience and tenure in the previous job, and the log difference in the national unemployment rate.

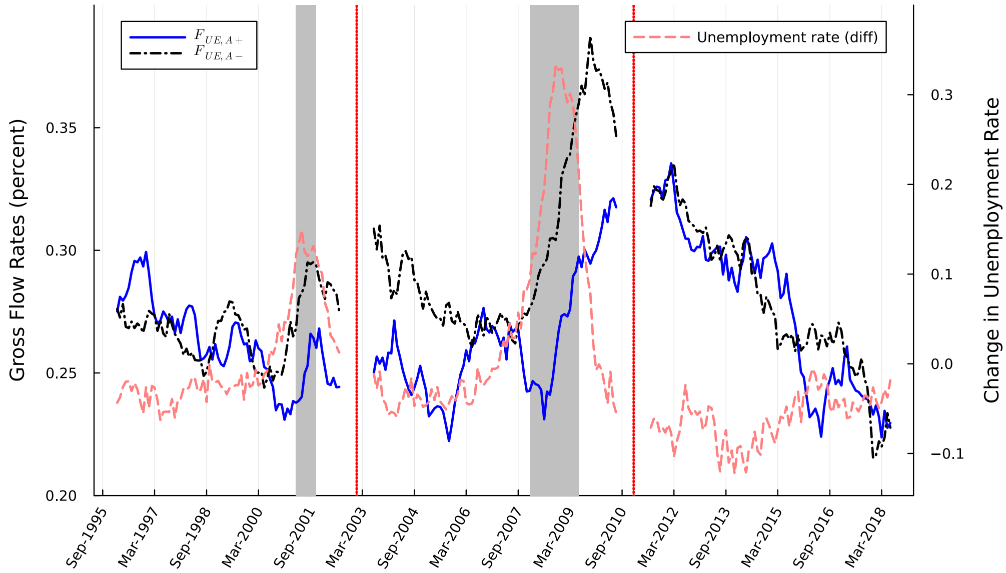

Figure E.20: Gross flow rates into and out of attractive occupations via UE transitions. Gray areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

Similarly, let $F_{UE,A+} = UE_{NA}/E_A$ and $F_{UE,A-} = UE_{AN}/E_A$ as the gross inflow and outflow rates for attractive occupations via UE transitions, respectively. Figure E.20 shows the time series of these flow rates. At the onset of the Great Recession, gross inflows into attractive occupations initially declined, whereas gross outflows increased. Later in the recession, both gross flow rates start increasing. A similar pattern is present in the 2001 recession, although this divergence between inflows and outflows starts a few months before the recession begins.

F Alternative classifications

In this section, we perform robustness analysis of the main findings under alternative occupation classifications. We focus exclusively on attractive occupations, as the corresponding time series for nonattractive occupations exhibit the same cyclical patterns but with opposite signs and different magnitudes.

F.1 Wages

As a robustness check, we examine whether our main results are sensitive to alternative definitions of attractive occupations based on wages. We consider two wage-based classifications: first, occupations with median wages above the national median; second, occupations satisfying either our baseline net inflow criterion or the high wage criterion.

Figure F.21 shows net inflow rates into attractive occupations via EE transitions over time, where attractive occupations are defined based on the wage criterion only. The cyclical pattern is less clear under this classification. Noting from Table 1, the wage-only criterion places office and administrative support, production, and healthcare support occupations in the nonattractive category. While these occupational groups pay below the national median wage, they may still be attractive destinations for workers employed in other low-paying occupations, such as sales and related, and personal care and service occupations. Figure A.11 indeed shows substantial inflows via EE transitions into these relatively low-wage occupations from other low-paying occupations.

Similarly, Figure F.22 shows net inflow rates into attractive occupations via EE transitions, where attractive occupations are defined by combining the net inflow and wage criteria. The resulting series is very similar to the series in Figure 1. This outcome occurs because the additional wage criterion re-labels only three occupation categories—legal; education, training, and library; and protective service—which already exhibit negative but low inflow rates. Since these occupations contribute minimally to overall EE transition flows, their reclassification has a negligible impact on the aggregate pattern.

Figures F.23 and F.24 show the net inflow rates via UE transitions under the alternative wage-based classifications. The interpretations are similar to those for the EE transitions.

F.2 3-Digit Occupations

Grouping occupations at the two-digit SOC level contains multiple skill levels and may potentially group different skill levels together. In this section, we group occupations at the three-digit SOC level and measure net inflows via EE transitions to

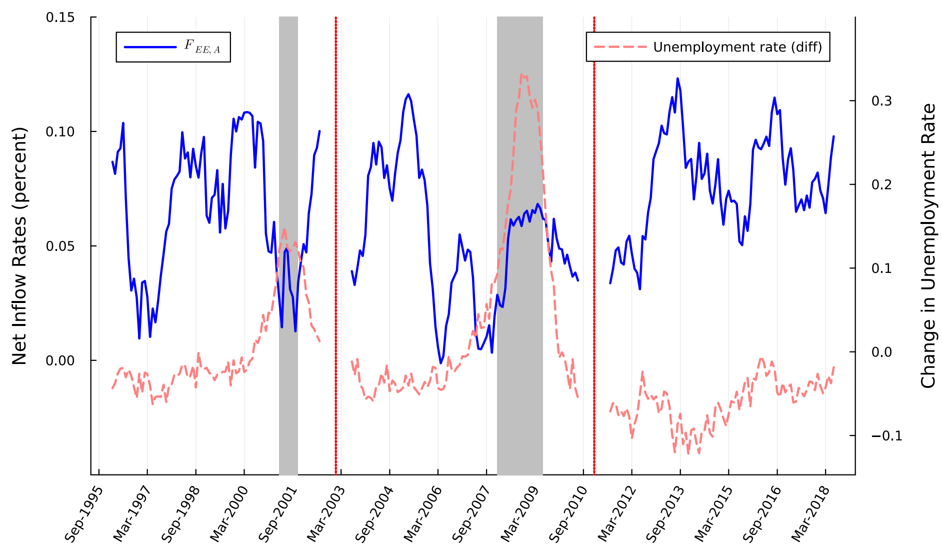

Figure F.21: Net inflow rates into attractive occupations via EE transitions. Attractive occupations are defined as those with a median wage above the national median. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

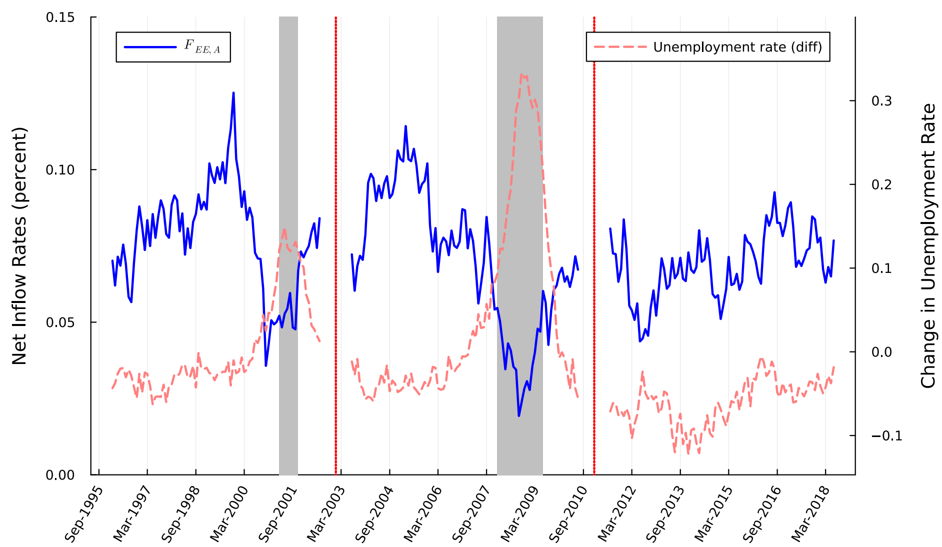

Figure F.22: Net inflow rates into attractive occupations via EE transitions. Attractive occupations are defined as those with a median wage above the national median or positive inflow rates. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

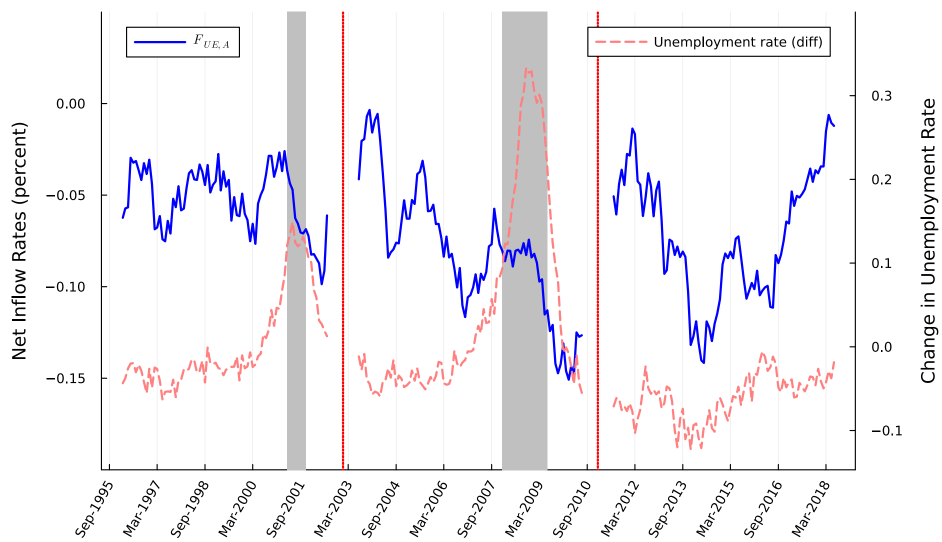

Figure F.23: Net inflow rates into attractive occupations via UE transitions. Attractive occupations are defined as those with a median wage above the national median. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

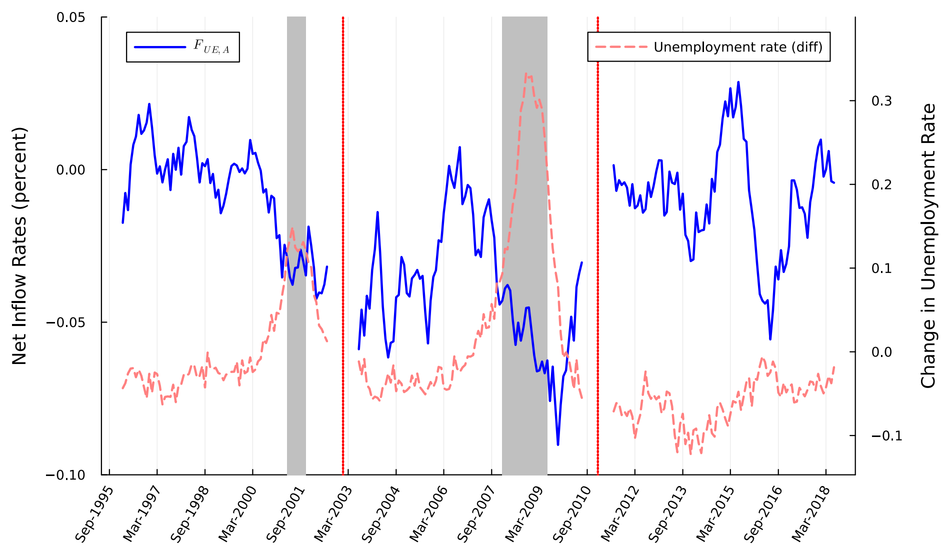

Figure F.24: Net inflow rates into attractive occupations via UE transitions. Attractive occupations are defined as those with a median wage above the national median or positive inflow rates. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

label them as attractive and nonattractive. Recall that the IPUMS-CPS harmonized variable (OCC2010) is based on the 2010 Census occupational classification. To obtain a mapping from this variable to the 2010 SOC classification system, we used the crosswalk table provided by the Census that maps the 2010 Census classification (OCC variable in IPUMS-CPS) to the 2010 SOC classification. We then matched the original occupation variable (OCC) to the corresponding harmonized variable in IPUMS-CPS (OCC2010).[26]

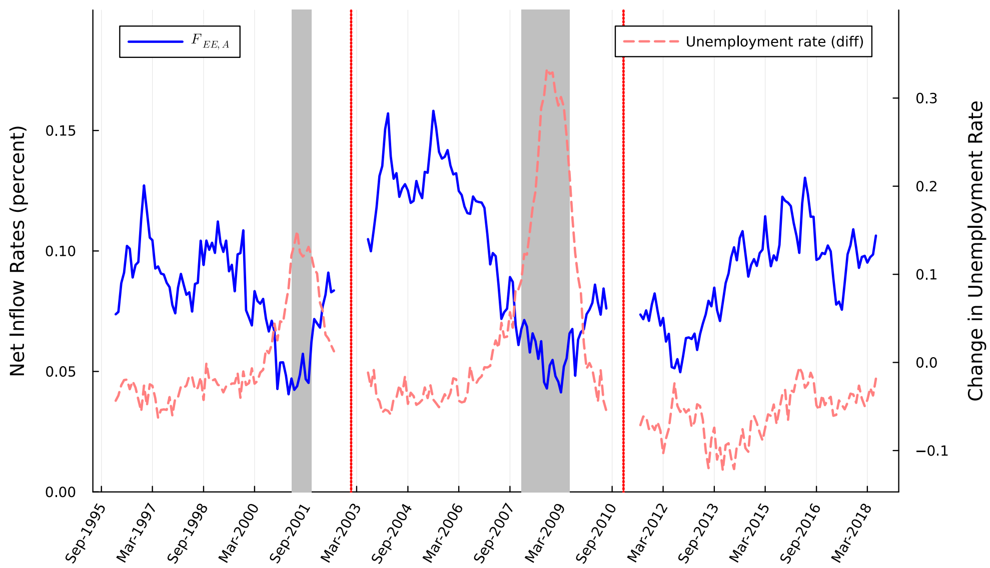

Figure F.25: Net inflow rate into attractive occupations via EE transitions using 3-digit SOC classification. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

The resulting series for net inflows via EE and UE transitions are displayed in Figures F.25 and F.26 for attractive occupations, respectively. Overall, both types of transitions exhibit more pronounced cyclicality compared to the corresponding figures in the main text that are based on 2-digit occupations.

[26]To perform this mapping, we utilized the March supplement from 2011 to ensure coverage of all occupations.

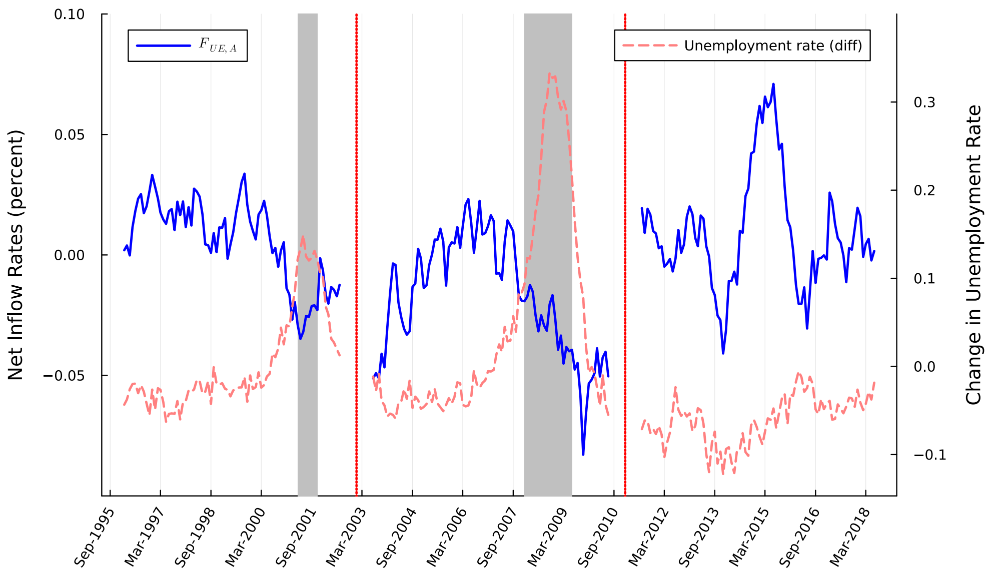

Figure F.26: Net inflow rate into attractive occupations via UE transitions using 3-digit SOC classification. Gray areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

F.3 Routine Manual Occupations

Although there is an extensive literature on job polarization documenting the decline in RM occupations (Acemoglu and Autor (2011)), we find that these occupations (construction and extraction; installation, maintenance, and repair; and production) have a positive net inflow via EE transitions. Except for production, the median wages for these occupations are also above the national median. In this subsection, we classify these occupations as a separate category to better understand their role in driving the net inflow rates in the main text.

Figure F.27 shows time series plots of net inflows from nonattractive occupations to RM occupations and other attractive occupations via EE transitions. Overall, the net inflow into RM occupations via EE is also positive and drops during both recessions. This pattern is similar to the net inflows into other attractive occupations, although the series for RM is noisier. This outcome suggests that the net inflows into attractive occupations via EE transitions are not driven by declining RM occupations.

However, net inflows via UE transitions differ substantially between RM and other attractive occupations. These series are shown in Figure F.28. The inflow rates from nonattractive to RM occupations via UE transitions are negative on average and larger in absolute value than those for other attractive occupations. Moreover, they decline sharply during recessions. In particular, the net outflows from RM occupations to nonattractive occupations via UE transitions during the Great Recession far exceed the average net inflow to RM occupations from nonattractive occupations. Overall, these findings suggest that the cyclicality in the UE-related transition is largely influenced by the behavior of net inflows into RM occupations.

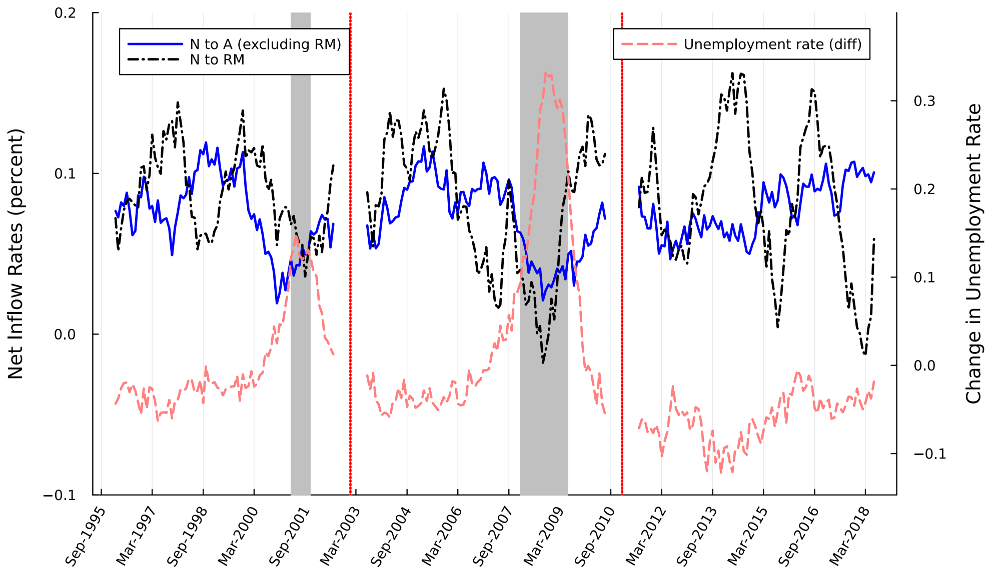

Figure F.27: Net inflow rate from nonattractive to RM and other attractive occupations via EE transitions. Gray areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

F.4 Entry Barriers

Entry into some occupations may be limited due to certificate and licensing requirements. Such restrictions would limit inflows (and potentially outflows) into these occupations, thereby complicating our categorization of occupations as attractive and nonattractive. In this subsection, we incorporate IPUMS-CPS data on certificate and licensing requirements to further group attractive and nonattractive occupations

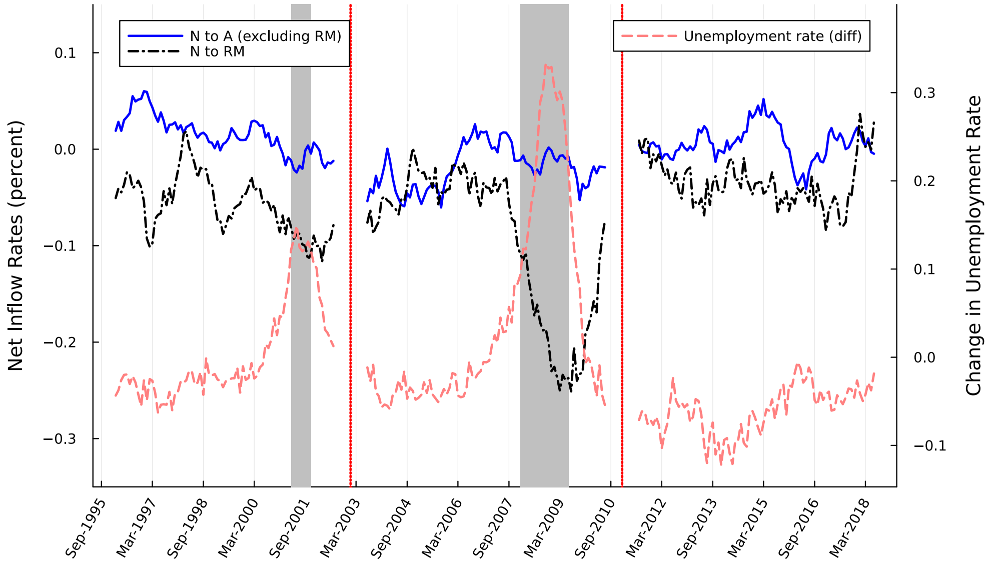

Figure F.28: Net inflow rate from nonattractive to RM and other attractive occupations via UE transitions. Gray areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

as high vs. low entry barrier occupations. Our goal is to isolate major occupation groups where there is greater potential for inflow from different occupations due to lower entry barriers.

The questions on certificate and licensing requirements are available in the basic monthly file only after 2016. To be consistent with the time frame of the main analysis, we use data through the end of 2018. Respondents are asked to indicate whether they have a professional certification or a state or industry license. In our sample, 24.76% report having such credentials. In Table 9 below, we report percentages by occupation. We classify occupations as having a high entry barrier if their occupation-level average exceeds the sample average. The differences are shown in the second column. Among nonattractive occupations, certificate and licensing requirements are particularly high in legal, education, protective services, and personal care and services.

We then construct net inflows from nonattractive to attractive occupations that have low entry barriers. The resulting series are displayed in Figures F.29 and F.30 for transitions via EE and UE, respectively. Inspection of these figures reveals that

Occupation title	Percentage	Difference from Sample Average
Management	0.222	−0.025
Business and financial operations	0.232	−0.016
Computer and mathematical	0.140	−0.107
Architecture and engineering	0.264	0.017
Life, physical, and social science	0.255	0.008
Community and social services	0.382	0.134
Legal	0.661	0.413
Education, training, and library	0.535	0.288
Arts, design, entertainment, sports, and media	0.112	−0.136
Healthcare practitioners and technical	0.771	0.524
Healthcare support	0.500	0.252
Protective service	0.366	0.119
Food preparation and serving related	0.076	−0.172
Building and grounds cleaning and maintenance	0.079	−0.168
Personal care and service	0.299	0.051
Sales and related	0.156	−0.092
Office and administrative support	0.095	−0.153
Farming, fishing, and forestry	0.098	−0.150
Construction and extraction	0.205	−0.043
Installation, maintenance, and repair	0.233	−0.014
Production	0.105	−0.143
Transportation and material moving	0.218	−0.030

Table 9: Certificate and Licensing Requirements

the series are very similar to Figures 1 and 3 in the main text.

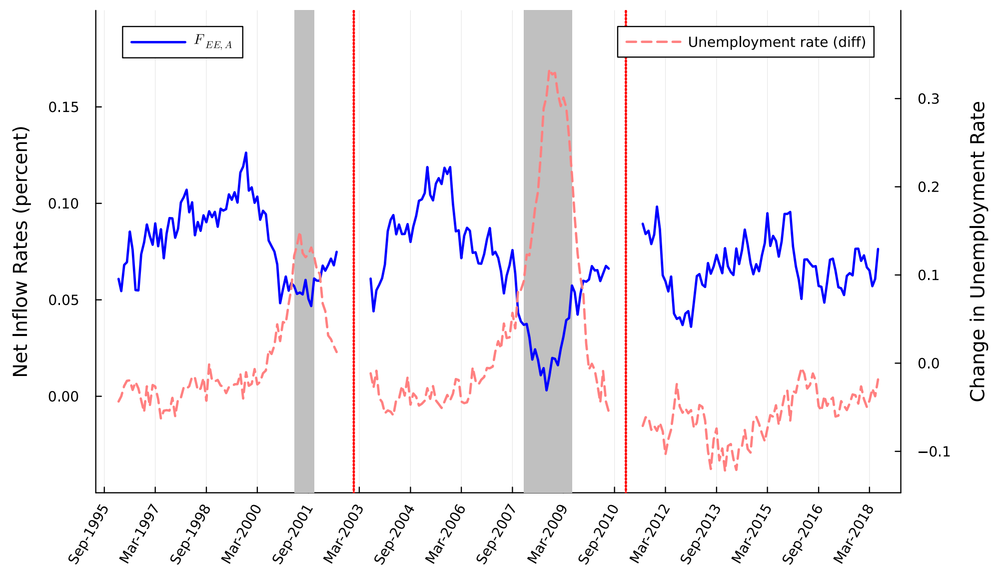

Figure F.29: Net inflow rate from nonattractive to attractive occupations via EE transitions where high entry barrier occupations are excluded. Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

F.5 EE rates: Adjustment

In a recent paper, Fujita et al. (2024) show that starting around 2008, there are many missing responses in CPS monthly data to the question of whether the respondent works for the same employer (IPUMS-CPS variable "EMPSAME"). They further show that these missing responses introduce a systematic bias in measuring EE transitions and propose an imputation methodology to correct for the bias in measuring EE rates. The resulting EE rates with imputed data dramatically differ from the series omitting the observations with missing values.

This data limitation may potentially impact how we measure net inflow rates and, consequently, the classification of occupations as nonattractive and attractive. To address these concerns, we obtained their data with imputed values for EE transitions and calculated the net inflow rates using this data.[27] In Table 10, we report the net

[27]We use the key variables that uniquely identify individuals to match their micro-level data with

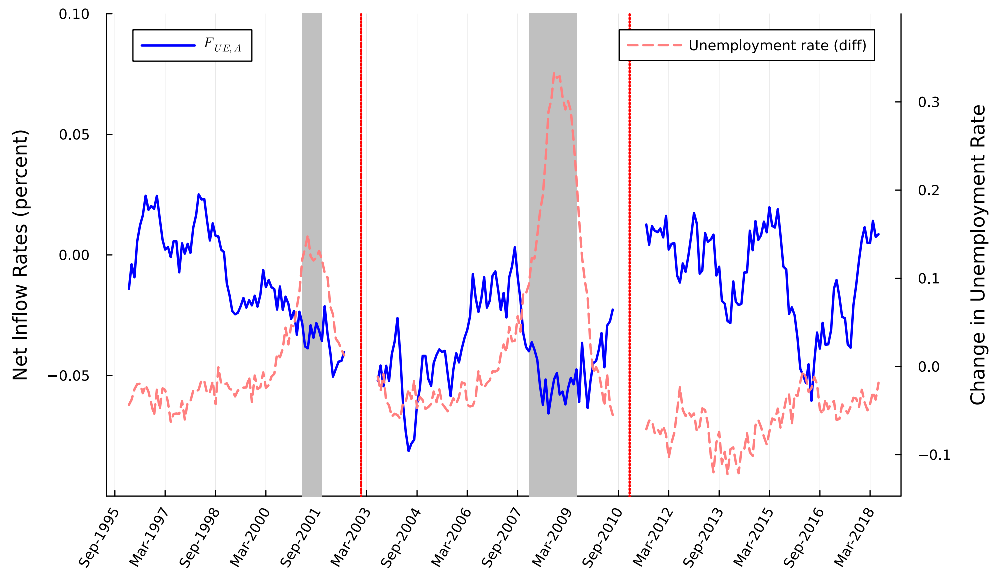

Figure F.30: Net inflow rate from nonattractive to attractive occupations via UE transitions where high entry barrier occupations are excluded. Gray areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.

inflow rates via EE transitions by occupation together with the baseline case where we drop observations with the missing values in the "EMPSAME" variable. Overall, the changes in net inflow rates via EE transitions are small when using the data with imputed values. Importantly, it does not change our classification with regard to attractive and nonattractive occupations.

We then move on to plotting net inflow rates via EE transitions over time using imputed data. The resulting series is shown in Figure F.31. The cyclical patterns are virtually unchanged, and our results in the main text are robust to this measurement correction.

imputed data to ours.

	Net inflow rates via EE	
Occupation title	Baseline	Imputed
Management	6.479	8.546
Business and financial operations	7.718	8.702
Computer and mathematical	6.576	5.330
Architecture and engineering	9.844	8.809
Life, physical, and social science	21.838	19.001
Community and social services	12.371	11.104
Legal	−0.653	−1.360
Education, training, and library	−6.286	−4.136
Arts, design, entertainment, sports, and media	4.575	3.794
Healthcare practitioners and technical	4.009	5.287
Healthcare support	16.456	14.328
Protective service	−1.156	−1.818
Food preparation and serving related	−32.974	−32.231
Building and grounds cleaning and maintenance	−9.725	−11.980
Personal care and service	−6.125	−6.952
Sales and related	−20.624	−21.711
Office and administrative support	7.061	9.103
Farming, fishing, and forestry	−9.084	−10.422
Construction and extraction	8.860	1.743
Installation, maintenance, and repair	9.747	9.659
Production	12.094	14.521
Transportation and material moving	2.200	1.010

Table 10: Net Inflow Rates by Occupation: The Baseline column reproduces values from Table 1. The Imputed column applies the imputation procedure described in Fujita et al. (2024).

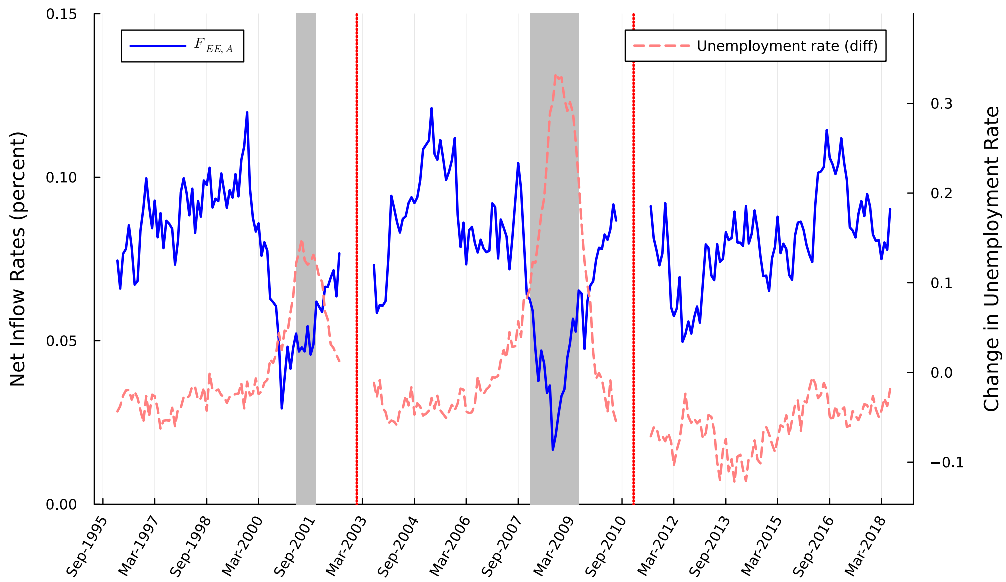

Figure F.31: Net inflow rates from nonattractive to attractive occupations via EE transitions using imputed data from Fujita et al. (2024). Gray shaded areas correspond to NBER recessions, and vertical red-dotted lines correspond to changes in SOC. The changes in the national unemployment rate are measured on the right axis. The monthly series is smoothed with a 12-month moving average.